Trust Region Policy Optimization (TRPO): A Formal Mathematical Treatment

Mathematical Foundations and Complete Proofs

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Abstract

This document provides a comprehensive mathematical treatment of Trust Region Policy Optimization (TRPO), including rigorous proofs of convergence, sample complexity analysis, detailed derivation of the loss function and its surrogate forms, neural network architecture considerations, and complete backpropagation derivations. We present TRPO as a theoretically principled algorithm for policy optimization in reinforcement learning with formal guarantees on monotonic improvement.

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1 Introduction and Motivation

Trust Region Policy Optimization (TRPO) addresses a fundamental challenge in policy gradient methods: how to take the largest possible improvement step without causing performance collapse. While vanilla policy gradient methods use fixed step sizes and suffer from sample inefficiency, TRPO provides a principled approach to adaptive step sizing based on trust region constraints.

1.1 The Policy Improvement Problem

Given a parameterized policy $\pi_\theta(a|s)$, we seek to maximize the expected return:

$$\eta(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t, a_t)\right] \tag{1}$$

The challenge is determining how to update θ to θ' such that $\eta(\theta') \geq \eta(\theta)$ with high confidence.

2 Theoretical Foundations

2.1 Policy Performance Difference

Definition 2.1 (Advantage Function)**.** The advantage function measures how much better an action is compared to the average action under policy π:

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s) \tag{2}$$

where $Q^\pi(s, a) = \mathbb{E}_\pi[\sum_{t=0}^{\infty}\gamma^t r_t|s_0 = s, a_0 = a]$ and $V^\pi(s) = \mathbb{E}_{a\sim\pi}[Q^\pi(s, a)]$.

Theorem 2.2 (Policy Improvement Bound - Kakade & Langford 2002)**.** *For any policies π and $\tilde{\pi}$, the difference in their performance can be expressed as:*

$$\eta(\tilde{\pi}) - \eta(\pi) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\tilde{\pi}}}\left[\mathbb{E}_{a\sim\tilde{\pi}(\cdot|s)}[A^\pi(s, a)]\right] \tag{3}$$

where $d^{\tilde{\pi}}(s) = (1-\gamma)\sum_{t=0}^{\infty}\gamma^t P(s_t = s|\tilde{\pi})$ is the discounted state visitation frequency.

Proof. This proof proceeds in several steps, using a telescoping argument combined with the Bellman equation structure.

Step 1: Initial setup.

The performance of policy $\tilde{\pi}$ is:

$$\eta(\tilde{\pi}) = \mathbb{E}_{\tau\sim\tilde{\pi}}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t, a_t)\right] = \mathbb{E}_{s_0\sim\rho_0}\left[V^{\tilde{\pi}}(s_0)\right] \tag{4}$$

where ρ_0 is the initial state distribution and $V^{\tilde{\pi}}(s_0)$ is the value function under policy $\tilde{\pi}$.

Step 2: Bellman decomposition at time $t = 0$.

We begin by decomposing $V^{\tilde{\pi}}(s_0)$. Using the definition of the advantage function:

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s) \tag{5}$$

we can write:

$$Q^\pi(s, a) = A^\pi(s, a) + V^\pi(s) \tag{6}$$

Now, the value under $\tilde{\pi}$ can be expressed as:

$$V^{\tilde{\pi}}(s_0) = \mathbb{E}_{a_0 \sim \tilde{\pi}(\cdot|s_0)} \left[Q^{\tilde{\pi}}(s_0, a_0) \right] \tag{7}$$
$$= \mathbb{E}_{a_0 \sim \tilde{\pi}} \left[r(s_0, a_0) + \gamma \mathbb{E}_{s_1 \sim P(\cdot|s_0, a_0)} \left[V^{\tilde{\pi}}(s_1) \right] \right] \tag{8}$$

The key insight is to relate $Q^{\tilde{\pi}}$ to quantities involving π. We add and subtract $V^{\pi}(s_0)$:

$$V^{\tilde{\pi}}(s_0) - V^{\pi}(s_0) = \mathbb{E}_{a_0 \sim \tilde{\pi}} \left[Q^{\tilde{\pi}}(s_0, a_0) \right] - V^{\pi}(s_0) \tag{9}$$
$$= \mathbb{E}_{a_0 \sim \tilde{\pi}} \left[Q^{\tilde{\pi}}(s_0, a_0) - V^{\pi}(s_0) \right] \tag{10}$$

Now we use a crucial observation: we can replace $Q^{\tilde{\pi}}(s_0, a_0)$ with $Q^{\pi}(s_0, a_0)$ plus a correction term. Specifically:

$$Q^{\tilde{\pi}}(s_0, a_0) = r(s_0, a_0) + \gamma \mathbb{E}_{s_1} \left[V^{\tilde{\pi}}(s_1) \right] \tag{11}$$
$$Q^{\pi}(s_0, a_0) = r(s_0, a_0) + \gamma \mathbb{E}_{s_1} \left[V^{\pi}(s_1) \right] \tag{12}$$

Therefore:

$$Q^{\tilde{\pi}}(s_0, a_0) = Q^{\pi}(s_0, a_0) + \gamma \mathbb{E}_{s_1} \left[V^{\tilde{\pi}}(s_1) - V^{\pi}(s_1) \right] \tag{13}$$

Step 3: First decomposition.
Substituting back:

$$V^{\tilde{\pi}}(s_0) - V^{\pi}(s_0) = \mathbb{E}_{a_0 \sim \tilde{\pi}} \left[Q^{\pi}(s_0, a_0) - V^{\pi}(s_0) \right] \tag{14}$$
$$+ \gamma \mathbb{E}_{a_0 \sim \tilde{\pi}} \mathbb{E}_{s_1} \left[V^{\tilde{\pi}}(s_1) - V^{\pi}(s_1) \right] \tag{15}$$
$$= \mathbb{E}_{a_0 \sim \tilde{\pi}} \left[A^{\pi}(s_0, a_0) \right] + \gamma \mathbb{E}_{a_0 \sim \tilde{\pi}, s_1} \left[V^{\tilde{\pi}}(s_1) - V^{\pi}(s_1) \right] \tag{16}$$

This is equation (6) from the original proof. Let me now explain this in more detail.
Step 4: Understanding equation (6).
Equation (6) states:

$$V^{\tilde{\pi}}(s_0) - V^{\pi}(s_0) = \mathbb{E}_{a_0 \sim \tilde{\pi}} [A^{\pi}(s_0, a_0)] + \mathbb{E}_{a_0 \sim \tilde{\pi}, s_1} \left[\gamma (V^{\tilde{\pi}}(s_1) - V^{\pi}(s_1)) \right] \tag{17}$$

This has a beautiful interpretation:

- The **first term** $\mathbb{E}_{a_0 \sim \tilde{\pi}}[A^{\pi}(s_0, a_0)]$ captures the immediate advantage of taking actions according to $\tilde{\pi}$ instead of π at state s_0, evaluated using the advantage function of policy π.
- The **second term** $\gamma \mathbb{E}_{a_0 \sim \tilde{\pi}, s_1}[V^{\tilde{\pi}}(s_1) - V^{\pi}(s_1)]$ represents the discounted future value difference, starting from the next state s_1 which is reached by taking action $a_0 \sim \tilde{\pi}$ and transitioning according to the dynamics.

Step 5: Recursive expansion (telescoping).
The key insight is that the second term has the same form as the left-hand side, just at state s_1 instead of s_0. We can therefore apply the same decomposition recursively:
At time $t = 1$:

$$V^{\tilde{\pi}}(s_1) - V^{\pi}(s_1) = \mathbb{E}_{a_1 \sim \tilde{\pi}}[A^{\pi}(s_1, a_1)] + \gamma \mathbb{E}_{a_1 \sim \tilde{\pi}, s_2} \left[V^{\tilde{\pi}}(s_2) - V^{\pi}(s_2) \right] \tag{18}$$

Substituting this into our equation from Step 3:

$$V^{\tilde{\pi}}(s_0) - V^{\pi}(s_0) = \mathbb{E}_{a_0 \sim \tilde{\pi}}[A^{\pi}(s_0, a_0)] \tag{19}$$
$$+ \gamma \mathbb{E}_{a_0, s_1, a_1 \sim \tilde{\pi}} \left[A^{\pi}(s_1, a_1) \right] \tag{20}$$
$$+ \gamma^2 \mathbb{E}_{a_0, s_1, a_1, s_2 \sim \tilde{\pi}} \left[V^{\tilde{\pi}}(s_2) - V^{\pi}(s_2) \right] \tag{21}$$

Continuing this process for $t = 2, 3, \ldots$:

$$V^{\tilde{\pi}}(s_0) - V^{\pi}(s_0) = \mathbb{E}_{a_0 \sim \tilde{\pi}}[A^{\pi}(s_0, a_0)] \tag{22}$$
$$+ \gamma \mathbb{E}_{s_1, a_1 \sim \tilde{\pi}}[A^{\pi}(s_1, a_1)] \tag{23}$$
$$+ \gamma^2 \mathbb{E}_{s_2, a_2 \sim \tilde{\pi}}[A^{\pi}(s_2, a_2)] \tag{24}$$
$$+ \gamma^3 \mathbb{E}_{s_3, a_3 \sim \tilde{\pi}}[A^{\pi}(s_3, a_3)] \tag{25}$$
$$+ \ldots \tag{26}$$

where all states s_t and actions a_t are generated by following policy $\tilde{\pi}$.

Step 6: Infinite sum convergence.

Since $\gamma < 1$ and rewards (hence advantages) are bounded, this infinite sum converges:

$$V^{\tilde{\pi}}(s_0) - V^{\pi}(s_0) = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim \tilde{\pi}}[A^{\pi}(s_t, a_t)] \tag{27}$$

More formally, when starting from s_0 and following $\tilde{\pi}$:

$$V^{\tilde{\pi}}(s_0) - V^{\pi}(s_0) = \mathbb{E}_{\tau \sim \tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi}(s_t, a_t) \right] \tag{28}$$

where $\tau = (s_0, a_0, s_1, a_1, \ldots)$ is a trajectory generated by $\tilde{\pi}$ starting from s_0.

Step 7: Averaging over initial states and introducing state visitation frequency.

Taking expectation over the initial state distribution ρ_0:

$$\eta(\tilde{\pi}) - \eta(\pi) = \mathbb{E}_{s_0 \sim \rho_0} \left[V^{\tilde{\pi}}(s_0) - V^{\pi}(s_0) \right] \tag{29}$$
$$= \mathbb{E}_{s_0 \sim \rho_0} \left[\mathbb{E}_{\tau \sim \tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi}(s_t, a_t) \right] \right] \tag{30}$$
$$= \mathbb{E}_{\tau \sim \tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi}(s_t, a_t) \right] \tag{31}$$

where the trajectory τ now includes sampling $s_0 \sim \rho_0$.

We can rewrite this by exchanging the sum and expectation:

$$\eta(\tilde{\pi}) - \eta(\pi) = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t \sim \tilde{\pi}, a_t \sim \tilde{\pi}(\cdot|s_t)}[A^{\pi}(s_t, a_t)] \tag{32}$$

Step 8: Introducing the normalized state visitation frequency.

Define the (unnormalized) probability of being in state s at time t when following $\tilde{\pi}$:

$$P(s_t = s|\tilde{\pi}) = \sum_{s_0} \rho_0(s_0) P(s_t = s|s_0, \tilde{\pi}) \tag{33}$$

The discounted state visitation frequency (normalized by $1 - \gamma$) is:

$$d^{\tilde{\pi}}(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P(s_t = s|\tilde{\pi}) \tag{34}$$

The normalization factor $(1 - \gamma)$ ensures this is a proper probability distribution:

$$\sum_s d^{\tilde{\pi}}(s) = (1 - \gamma) \sum_s \sum_{t=0}^{\infty} \gamma^t P(s_t = s|\tilde{\pi}) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \sum_s P(s_t = s|\tilde{\pi}) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t = 1 \tag{35}$$

Step 9: Final form.
Using the state visitation frequency:

$$\eta(\tilde{\pi}) - \eta(\pi) = \sum_{t=0}^{\infty} \gamma^t \sum_s P(s_t = s|\tilde{\pi}) \mathbb{E}_{a\sim\tilde{\pi}(\cdot|s)}[A^\pi(s,a)] \tag{36}$$

$$= \sum_s \left[\sum_{t=0}^{\infty} \gamma^t P(s_t = s|\tilde{\pi})\right] \mathbb{E}_{a\sim\tilde{\pi}(\cdot|s)}[A^\pi(s,a)] \tag{37}$$

$$= \frac{1}{1-\gamma} \sum_s \left[(1-\gamma)\sum_{t=0}^{\infty} \gamma^t P(s_t = s|\tilde{\pi})\right] \mathbb{E}_{a\sim\tilde{\pi}(\cdot|s)}[A^\pi(s,a)] \tag{38}$$

$$= \frac{1}{1-\gamma} \sum_s d^{\tilde{\pi}}(s) \mathbb{E}_{a\sim\tilde{\pi}(\cdot|s)}[A^\pi(s,a)] \tag{39}$$

$$= \frac{1}{1-\gamma} \mathbb{E}_{s\sim d^{\tilde{\pi}}}\left[\mathbb{E}_{a\sim\tilde{\pi}(\cdot|s)}[A^\pi(s,a)]\right] \tag{40}$$

This completes the proof. □

Remark 2.3 (Understanding Equation (6) and the Telescoping Structure). The key equation in Step 4 of the proof:

$$V^{\tilde{\pi}}(s_0) - V^\pi(s_0) = \mathbb{E}_{a_0\sim\tilde{\pi}}[A^\pi(s_0,a_0)] + \gamma \mathbb{E}_{s_1,a_0\sim\tilde{\pi}}\left[V^{\tilde{\pi}}(s_1) - V^\pi(s_1)\right] \tag{41}$$

is the foundation of the entire policy improvement theory. Let us examine why this equation is both natural and profound.

The Bellman Structure: This decomposition exploits the recursive structure of the Bellman equation. The value difference can be split into:

1. **Immediate advantage:** $\mathbb{E}_{a_0\sim\tilde{\pi}}[A^\pi(s_0,a_0)]$ measures how much better (or worse) it is to sample actions from $\tilde{\pi}$ at state s_0 compared to π, evaluated using π's advantage function.

2. **Propagated difference:** $\gamma\mathbb{E}[V^{\tilde{\pi}}(s_1) - V^\pi(s_1)]$ captures all future value differences, discounted by γ and weighted by the distribution over next states induced by $\tilde{\pi}$.

Why use A^π instead of $A^{\tilde{\pi}}$? A crucial aspect is that we evaluate actions using A^π, the advantage function of the *old* policy π, not the new policy $\tilde{\pi}$. This is essential because:

- A^π is *known* - we can estimate it from data collected under π
- $A^{\tilde{\pi}}$ is *unknown* - we haven't executed $\tilde{\pi}$ yet
- The relationship $Q^{\tilde{\pi}}(s,a) = Q^\pi(s,a) + \gamma\mathbb{E}_{s'}[V^{\tilde{\pi}}(s') - V^\pi(s')]$ allows us to factor out the difference in Q-functions

The Telescoping Mechanism: When we apply equation (6) recursively, we get:

$$\begin{aligned} V^{\tilde{\pi}}(s_0) - V^\pi(s_0) &= A^\pi(s_0,a_0) + \gamma[V^{\tilde{\pi}}(s_1) - V^\pi(s_1)] \\ &= A^\pi(s_0,a_0) + \gamma A^\pi(s_1,a_1) + \gamma^2[V^{\tilde{\pi}}(s_2) - V^\pi(s_2)] \\ &= A^\pi(s_0,a_0) + \gamma A^\pi(s_1,a_1) + \gamma^2 A^\pi(s_2,a_2) + \gamma^3[\ldots] \\ &= \sum_{t=0}^{\infty} \gamma^t A^\pi(s_t,a_t) \end{aligned}$$

where each $V^{\tilde{\pi}}(s_t) - V^\pi(s_t)$ term is "opened up" using equation (6), revealing the advantage at time t plus a residual term for time $t+1$. The $\gamma < 1$ ensures convergence.

Connection to Policy Gradient: When $\tilde{\pi} = \pi_{\theta+d\theta}$ is an infinitesimal perturbation of π_θ:

$$\begin{aligned}\eta(\theta + d\theta) - \eta(\theta) &\approx \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_\theta}}\left[\mathbb{E}_{a\sim\pi_{\theta+d\theta}}[A^{\pi_\theta}(s,a)]\right] \\ &\approx \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_\theta}}\left[\mathbb{E}_{a\sim\pi_\theta}[A^{\pi_\theta}(s,a)]\right] + d\theta^\top g \\ &= 0 + d\theta^\top g\end{aligned}$$

where $g = \nabla_\theta \eta(\theta)$ is the policy gradient. The first term vanishes because $\mathbb{E}_{a\sim\pi}[A^\pi(s,a)] = 0$ by definition of advantage.

Why This Matters for TRPO: The entire trust region approach is built on this identity. By showing that:

$$\eta(\tilde{\pi}) = \eta(\pi) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\tilde{\pi}}}[\mathbb{E}_{a\sim\tilde{\pi}}[A^\pi(s,a)]]$$

we see that improving $\eta(\tilde{\pi})$ requires making $\mathbb{E}_{a\sim\tilde{\pi}}[A^\pi(s,a)]$ positive in expectation over the state distribution $d^{\tilde{\pi}}$. The challenge is that $d^{\tilde{\pi}}$ depends on $\tilde{\pi}$, which is unknown. TRPO's key innovation is replacing $d^{\tilde{\pi}}$ with d^π (which we can estimate) and bounding the error introduced by this substitution using the KL divergence between $\tilde{\pi}$ and π.

2.2 Surrogate Objective with State Visitation Mismatch

The key challenge is that $d^{\tilde{\pi}}$ depends on the new policy $\tilde{\pi}$, making direct optimization intractable. TRPO addresses this by replacing $d^{\tilde{\pi}}$ with d^π.

Definition 2.4 (Surrogate Advantage)**.** Define the surrogate advantage:

$$L_\pi(\tilde{\pi}) = \mathbb{E}_{s\sim d^\pi}\left[\mathbb{E}_{a\sim\tilde{\pi}(\cdot|s)}[A^\pi(s,a)]\right] \tag{42}$$

Lemma 2.5 (Local Approximation Property)**.** *At $\tilde{\pi} = \pi$, we have:*

$$L_\pi(\pi) = 0, \quad \nabla_\theta L_\pi(\pi_\theta)\big|_{\theta=\theta_{old}} = \nabla_\theta \eta(\theta)\big|_{\theta=\theta_{old}} \tag{43}$$

Proof. We prove both parts using only Definition 2.2 above.

Part 1: Proving $L_\pi(\pi) = 0$.

Using Definition 2.2 with $\tilde{\pi} = \pi$:

$$L_\pi(\pi) = \mathbb{E}_{s\sim d^\pi}\left[\mathbb{E}_{a\sim\pi(\cdot|s)}[A^\pi(s,a)]\right] \tag{44}$$

Now, for any policy π and any state s, we have the fundamental property:

$$\mathbb{E}_{a\sim\pi(\cdot|s)}[A^\pi(s,a)] = 0 \tag{45}$$

Proof of this property: By definition of advantage:

$$\begin{aligned}\mathbb{E}_{a\sim\pi(\cdot|s)}[A^\pi(s,a)] &= \mathbb{E}_{a\sim\pi(\cdot|s)}[Q^\pi(s,a) - V^\pi(s)] &(46)\\ &= \mathbb{E}_{a\sim\pi(\cdot|s)}[Q^\pi(s,a)] - V^\pi(s) &(47)\\ &= V^\pi(s) - V^\pi(s) &(48)\\ &= 0 &(49)\end{aligned}$$

where we used that $V^\pi(s) = \mathbb{E}_{a\sim\pi}[Q^\pi(s,a)]$ by definition.

Therefore:

$$L_\pi(\pi) = \mathbb{E}_{s\sim d^\pi}[0] = 0 \tag{50}$$

Part 2: Proving $\nabla_\theta L_\pi(\pi_\theta)\big|_{\theta=\theta_{\mathbf{old}}} = \nabla_\theta \eta(\theta)\big|_{\theta=\theta_{\mathbf{old}}}$.

Step 1: Compute the gradient of L.

From Definition 2.2, with $\pi = \pi_{\theta_{\text{old}}}$ and $\tilde{\pi} = \pi_\theta$:

$$L_{\theta_{\text{old}}}(\theta) = \mathbb{E}_{s\sim d^{\theta_{\text{old}}}}\left[\mathbb{E}_{a\sim\pi_\theta}[A^{\theta_{\text{old}}}(s,a)]\right] \tag{51}$$

Taking the gradient with respect to θ:

$$\begin{aligned}\nabla_\theta L_{\theta_{\text{old}}}(\theta) &= \nabla_\theta \mathbb{E}_{s\sim d^{\theta_{\text{old}}}}\left[\mathbb{E}_{a\sim\pi_\theta}[A^{\theta_{\text{old}}}(s,a)]\right] &(52)\\ &= \mathbb{E}_{s\sim d^{\theta_{\text{old}}}}\left[\nabla_\theta \mathbb{E}_{a\sim\pi_\theta}[A^{\theta_{\text{old}}}(s,a)]\right] &(53)\end{aligned}$$

Note that the gradient passes through the outer expectation because $d^{\theta_{\text{old}}}$ does not depend on θ (it depends only on θ_{old}).

Step 2: Apply the log-derivative trick.

For the inner expectation:

$$\begin{aligned}\nabla_\theta \mathbb{E}_{a\sim\pi_\theta}[A^{\theta_{\text{old}}}(s,a)] &= \nabla_\theta \sum_a \pi_\theta(a|s) A^{\theta_{\text{old}}}(s,a) &(54)\\ &= \sum_a \nabla_\theta \pi_\theta(a|s) A^{\theta_{\text{old}}}(s,a) &(55)\\ &= \sum_a \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) A^{\theta_{\text{old}}}(s,a) &(56)\\ &= \mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) A^{\theta_{\text{old}}}(s,a)\right] &(57)\end{aligned}$$

where we used $\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \log \pi_\theta$.

Therefore:

$$\nabla_\theta L_{\theta_{\text{old}}}(\theta) = \mathbb{E}_{s\sim d^{\theta_{\text{old}}}}\left[\mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) A^{\theta_{\text{old}}}(s,a)\right]\right] \tag{58}$$

Step 3: Evaluate at $\theta = \theta_{\text{old}}$.

When $\theta = \theta_{\text{old}}$:

$$\nabla_\theta L_{\theta_{\text{old}}}(\theta)\Big|_{\theta=\theta_{\text{old}}} = \mathbb{E}_{s\sim d^{\theta_{\text{old}}}}\left[\mathbb{E}_{a\sim\pi_{\theta_{\text{old}}}}\left[\nabla_\theta \log \pi_\theta(a|s)\Big|_{\theta=\theta_{\text{old}}} A^{\theta_{\text{old}}}(s,a)\right]\right] \tag{59}$$

Step 4: Relate to the policy gradient.

Now we need to show this equals $\nabla_\theta \eta(\theta)\big|_{\theta=\theta_{\text{old}}}$.

Recall the policy gradient theorem (Sutton et al.):

$$\nabla_\theta \eta(\theta) = \mathbb{E}_{s\sim d^{\pi_\theta}}\left[\mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s,a)\right]\right] \tag{60}$$

We can equivalently write this using advantages:

$$\begin{aligned}\nabla_\theta \eta(\theta) &= \mathbb{E}_{s\sim d^{\pi_\theta}}\left[\mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s,a)\right]\right] &(61)\\ &= \mathbb{E}_{s\sim d^{\pi_\theta}}\left[\mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s)(A^{\pi_\theta}(s,a) + V^{\pi_\theta}(s))\right]\right] &(62)\\ &= \mathbb{E}_{s\sim d^{\pi_\theta}}\left[\mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) A^{\pi_\theta}(s,a)\right]\right] &(63)\\ &\quad + \mathbb{E}_{s\sim d^{\pi_\theta}}\left[\mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) V^{\pi_\theta}(s)\right]\right] &(64)\\ &= \mathbb{E}_{s\sim d^{\pi_\theta}}\left[\mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) A^{\pi_\theta}(s,a)\right]\right] &(65)\\ &\quad + \mathbb{E}_{s\sim d^{\pi_\theta}}\left[V^{\pi_\theta}(s)\mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s)\right]\right] &(66)\end{aligned}$$

The second term is zero because:

$$\mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s)\right] = \nabla_\theta \sum_a \pi_\theta(a|s) = \nabla_\theta 1 = 0 \tag{67}$$

Therefore:

$$\nabla_\theta \eta(\theta) = \mathbb{E}_{s\sim d^{\pi_\theta}} \left[\mathbb{E}_{a\sim\pi_\theta} \left[\nabla_\theta \log \pi_\theta(a|s) A^{\pi_\theta}(s,a)\right]\right] \tag{68}$$

Step 5: Evaluate at $\theta = \theta_{\text{old}}$.
At $\theta = \theta_{\text{old}}$, we have $\pi_\theta = \pi_{\theta_{\text{old}}}$, so:

$$\nabla_\theta \eta(\theta)\Big|_{\theta=\theta_{\text{old}}} = \mathbb{E}_{s\sim d^{\theta_{\text{old}}}} \left[\mathbb{E}_{a\sim\pi_{\theta_{\text{old}}}} \left[\nabla_\theta \log \pi_\theta(a|s)\Big|_{\theta=\theta_{\text{old}}} A^{\theta_{\text{old}}}(s,a)\right]\right] \tag{69}$$

Step 6: Compare with gradient of L.
From Step 3:

$$\nabla_\theta L_{\theta_{\text{old}}}(\theta)\Big|_{\theta=\theta_{\text{old}}} = \mathbb{E}_{s\sim d^{\theta_{\text{old}}}} \left[\mathbb{E}_{a\sim\pi_{\theta_{\text{old}}}} \left[\nabla_\theta \log \pi_\theta(a|s)\Big|_{\theta=\theta_{\text{old}}} A^{\theta_{\text{old}}}(s,a)\right]\right] \tag{70}$$

These are identical! Therefore:

$$\nabla_\theta L_{\theta_{\text{old}}}(\theta)\Big|_{\theta=\theta_{\text{old}}} = \nabla_\theta \eta(\theta)\Big|_{\theta=\theta_{\text{old}}} \tag{71}$$

This completes the proof using only our Definition 2.2. □

Remark 2.6 (Significance of the Local Approximation Property). This lemma is fundamental because it shows that:

1. **At the current policy**, the surrogate objective equals zero: $L_{\text{old}}(\text{old}) = 0$
2. **The gradients match**, so locally (for small updates), optimizing L is equivalent to optimizing η

This justifies using L as a surrogate: it has the same gradient as the true objective at the current policy, making it a valid local approximation for policy improvement.

2.3 Bounding the State Visitation Mismatch

Theorem 2.7 (Performance Difference Bound). *For any policies $\pi_{\theta_{old}}$ and $\pi_{\theta_{new}}$:*

$$\eta(\theta_{new}) \geq L_{\theta_{old}}(\theta_{new}) - \frac{4\epsilon\gamma}{(1-\gamma)^2}\alpha^2 \tag{72}$$

where $\epsilon = \max_s |\mathbb{E}_{a\sim\pi_{\theta_{new}}}[A^{\theta_{old}}(s,a)]|$ and $\alpha = \max_s KL(\pi_{\theta_{new}}(\cdot|s)\|\pi_{\theta_{old}}(\cdot|s))$.

Proof. From Theorem 2.2, we have:

$$\eta(\theta_{\text{new}}) = \eta(\theta_{\text{old}}) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\theta_{\text{new}}}} \left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]\right] \tag{73}$$

We can rewrite using $d^{\theta_{\text{old}}}$ instead:

$$= \eta(\theta_{\text{old}}) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\theta_{\text{old}}}} \left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]\right] \tag{74}$$

$$+ \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\theta_{\text{new}}}-d^{\theta_{\text{old}}}} \left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]\right] \tag{75}$$

Lemma 2.8 (Expectation Decomposition via Signed Measures). *For any function $f : \mathcal{S} \to \mathbb{R}$ and probability distributions P and Q over $\mathcal{S}$:*

$$\mathbb{E}_{s\sim P}[f(s)] = \mathbb{E}_{s\sim Q}[f(s)] + \mathbb{E}_{s\sim P-Q}[f(s)] \tag{76}$$

where $\mathbb{E}_{s\sim P-Q}[f(s)] := \sum_s (P(s)-Q(s))f(s)$ is the expectation with respect to the signed measure $(P-Q)$.

Proof. **Step 1: Definition of expectation.**
By definition of expectation under probability distribution P:

$$\mathbb{E}_{s\sim P}[f(s)] = \sum_{s\in\mathcal{S}} P(s)f(s) \tag{77}$$

Similarly for Q:

$$\mathbb{E}_{s\sim Q}[f(s)] = \sum_{s\in\mathcal{S}} Q(s)f(s) \tag{78}$$

Step 2: Signed measure definition.
Define the signed measure $\mu = P - Q$:

$$\mu(s) = P(s) - Q(s) \tag{79}$$

Note that μ is not a probability measure since:

- $\sum_s \mu(s) = \sum_s P(s) - \sum_s Q(s) = 1 - 1 = 0$
- $\mu(s)$ can be negative for some states s

The expectation with respect to this signed measure is:

$$\mathbb{E}_{s\sim P-Q}[f(s)] = \sum_{s\in\mathcal{S}} \mu(s)f(s) = \sum_{s\in\mathcal{S}} (P(s) - Q(s))f(s) \tag{80}$$

Step 3: Algebraic decomposition.
Expanding the signed measure expectation:

$$\begin{aligned}
\mathbb{E}_{s\sim P-Q}[f(s)] &= \sum_{s\in\mathcal{S}} (P(s) - Q(s))f(s) && (81)\\
&= \sum_{s\in\mathcal{S}} P(s)f(s) - \sum_{s\in\mathcal{S}} Q(s)f(s) && (82)\\
&= \mathbb{E}_{s\sim P}[f(s)] - \mathbb{E}_{s\sim Q}[f(s)] && (83)
\end{aligned}$$

Step 4: Rearranging.
Rearranging the above equation:

$$\mathbb{E}_{s\sim P}[f(s)] = \mathbb{E}_{s\sim Q}[f(s)] + \mathbb{E}_{s\sim P-Q}[f(s)] \tag{84}$$

This completes the proof. □

Remark 2.9 (Understanding Signed Measures in RL). The signed measure $(P - Q)$ appears naturally when comparing two policies:

1. Intuition: The term $\mathbb{E}_{s\sim P-Q}[f(s)]$ represents the *difference* in expected values when sampling from P versus Q. States where $P(s) > Q(s)$ contribute positively (we visit them more under P), while states where $P(s) < Q(s)$ contribute negatively (we visit them less under P).

2. In TRPO context: With $P = d^{\theta_{\text{new}}}$, $Q = d^{\theta_{\text{old}}}$, and $f(s) = \mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]$:

$$\underbrace{\mathbb{E}_{s\sim d^{\theta_{\text{new}}}}[f(s)]}_{\text{True improvement}} = \underbrace{\mathbb{E}_{s\sim d^{\theta_{\text{old}}}}[f(s)]}_{\text{Surrogate (easy to estimate)}} + \underbrace{\mathbb{E}_{s\sim d^{\theta_{\text{new}}}-d^{\theta_{\text{old}}}}[f(s)]}_{\text{Error term (must be bounded)}} \tag{85}$$

3. Why this decomposition is useful:

- $\mathbb{E}_{s\sim d^{\theta_{\text{old}}}}[\cdots]$ can be estimated from samples collected under the old policy
- $\mathbb{E}_{s\sim d^{\theta_{\text{new}}}}[\cdots]$ cannot be estimated (we haven't executed the new policy yet)

- The error term $\mathbb{E}_{s\sim d^{\theta_{\text{new}}}-d^{\theta_{\text{old}}}}[\cdots]$ can be bounded using KL divergence between policies

4. Connection to importance sampling: This is analogous to importance sampling, but at the level of state distributions rather than individual transitions. In importance sampling, we have:

$$\mathbb{E}_{x\sim p}[f(x)] = \mathbb{E}_{x\sim q}\left[\frac{p(x)}{q(x)}f(x)\right] \tag{86}$$

Our decomposition is more subtle because we're dealing with the induced state distributions of entire Markov chains.

5. Geometric interpretation: If we think of probability distributions as vectors in $\mathbb{R}^{|\mathcal{S}|}$ (with entries summing to 1), then $P - Q$ is a vector that:

- Points in the direction of the difference
- Has components summing to zero (it's in the hyperplane orthogonal to the all-ones vector)
- Can be decomposed into positive and negative parts: $(P-Q)^+ = \max(P-Q, 0)$ and $(P-Q)^- = \max(Q-P, 0)$

The total variation distance is exactly $\|(P-Q)^+\|_1 = \|(P-Q)^-\|_1 = \frac{1}{2}\|P-Q\|_1$.

Applying Lemma 2.10 to TRPO:

In our case, with:

- $P = d^{\theta_{\text{new}}}$ (state visitation under new policy)
- $Q = d^{\theta_{\text{old}}}$ (state visitation under old policy)
- $f(s) = \mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]$ (expected advantage at state s)

We obtain:

$$\mathbb{E}_{s\sim d^{\theta_{\text{new}}}}\left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]\right] = \mathbb{E}_{s\sim d^{\theta_{\text{old}}}}\left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]\right] \tag{87}$$

$$+ \mathbb{E}_{s\sim d^{\theta_{\text{new}}}-d^{\theta_{\text{old}}}}\left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]\right] \tag{88}$$

This is exactly equations (53)-(54) in the main proof.

Justification of the distribution decomposition:

The identity used here is a fundamental result about expectations and signed measures. We prove it rigorously:

Lemma 2.10 (Expectation Decomposition via Signed Measures)**.** *For any two probability distributions P and Q on a finite or countable space $\mathcal{S}$, and any measurable function $f : \mathcal{S} \to \mathbb{R}$:*

$$\mathbb{E}_{s\sim P}[f(s)] = \mathbb{E}_{s\sim Q}[f(s)] + \mathbb{E}_{s\sim P-Q}[f(s)] \tag{89}$$

where the notation $\mathbb{E}_{s\sim P-Q}[f(s)]$ is defined as:

$$\mathbb{E}_{s\sim P-Q}[f(s)] := \sum_{s\in\mathcal{S}}[P(s)-Q(s)]f(s) \tag{90}$$

Proof. This follows directly from linearity of expectation and the definition of expectation as a weighted sum.

Step 1: Expand the left-hand side.

$$\mathbb{E}_{s\sim P}[f(s)] = \sum_{s\in\mathcal{S}} P(s)f(s) \tag{91}$$

Step 2: Add and subtract $Q(s)$ inside the sum.

$$\mathbb{E}_{s\sim P}[f(s)] = \sum_{s\in\mathcal{S}} P(s)f(s) \tag{92}$$
$$= \sum_{s\in\mathcal{S}} [Q(s) + (P(s) - Q(s))]f(s) \tag{93}$$
$$= \sum_{s\in\mathcal{S}} Q(s)f(s) + \sum_{s\in\mathcal{S}} [P(s) - Q(s)]f(s) \tag{94}$$
$$= \mathbb{E}_{s\sim Q}[f(s)] + \mathbb{E}_{s\sim P-Q}[f(s)] \tag{95}$$

Step 3: Interpretation of the signed measure.

The term $\mathbb{E}_{s\sim P-Q}[f(s)]$ is an expectation with respect to a *signed measure* $\mu = P - Q$, where:

$$\mu(s) = P(s) - Q(s) \tag{96}$$

Key properties:

- μ is not a probability measure (can be negative)
- $\sum_s \mu(s) = \sum_s P(s) - \sum_s Q(s) = 1 - 1 = 0$ (zero total mass)
- $\mu(s) > 0$ for states where P puts more mass than Q
- $\mu(s) < 0$ for states where Q puts more mass than P

This identity is a direct consequence of the linearity of integration/summation. □

Remark 2.11 (Understanding the Signed Measure Term). The signed measure expectation $\mathbb{E}_{s\sim P-Q}[f(s)]$ can be decomposed further to understand its meaning:

1. Positive and negative parts: Define:

$$\mu^+(s) = \max(0, P(s) - Q(s)) = P(s) - \min(P(s), Q(s)) \tag{97}$$
$$\mu^-(s) = \max(0, Q(s) - P(s)) = Q(s) - \min(P(s), Q(s)) \tag{98}$$

Then $\mu = \mu^+ - \mu^-$ and:

$$\mathbb{E}_{s\sim P-Q}[f(s)] = \sum_s [P(s) - Q(s)]f(s) \tag{99}$$
$$= \sum_s \mu^+(s)f(s) - \sum_s \mu^-(s)f(s) \tag{100}$$

2. Interpretation as difference in expectations:

$$\mathbb{E}_{s\sim P-Q}[f(s)] = \sum_s P(s)f(s) - \sum_s Q(s)f(s) \tag{101}$$
$$= \mathbb{E}_{s\sim P}[f(s)] - \mathbb{E}_{s\sim Q}[f(s)] \tag{102}$$

This is exactly the difference we want to bound!

3. Bounding via total variation:

$$|\mathbb{E}_{s\sim P-Q}[f(s)]| = \left|\sum_s [P(s) - Q(s)]f(s)\right| \tag{103}$$
$$\leq \sum_s |P(s) - Q(s)| \cdot |f(s)| \tag{104}$$
$$\leq \max_s |f(s)| \cdot \sum_s |P(s) - Q(s)| \tag{105}$$
$$= \|f\|_\infty \cdot \mathrm{TV}(P, Q) \cdot 2 \tag{106}$$

where $\mathrm{TV}(P, Q) = \frac{1}{2}\sum_s |P(s) - Q(s)|$ is the total variation distance.

4. Application to TRPO: In our setting:

- $P = d^{\theta_{\text{new}}}$ (state distribution under new policy)
- $Q = d^{\theta_{\text{old}}}$ (state distribution under old policy)
- $f(s) = \mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s, a)]$ (expected advantage at state s)
- $\|f\|_\infty \leq \epsilon$ (bounded advantage assumption)

Therefore:

$$\left|\frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\theta_{\text{new}}}-d^{\theta_{\text{old}}}}[f(s)]\right| \leq \frac{2\epsilon}{1-\gamma}\mathrm{TV}(d^{\theta_{\text{new}}}, d^{\theta_{\text{old}}}) \tag{107}$$

The entire TRPO proof hinges on bounding this TV distance using the KL divergence between policies!

Lemma 2.12 (Surrogate Objective with Scaling Factor)**.** *The surrogate objective from Definition 2.2, when expressed with the* $(1-\gamma)$ *normalization from the policy improvement theorem, satisfies:*

$$\frac{1}{1-\gamma}L_{\theta_{old}}(\theta_{new}) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\theta_{old}}}\left[\mathbb{E}_{a\sim\pi_{\theta_{new}}}[A^{\theta_{old}}(s, a)]\right] \tag{108}$$

Proof. This is simply Definition 2.2 multiplied by the constant factor $\frac{1}{1-\gamma}$. The factor arises from the policy improvement theorem (Theorem 2.2), which states:

$$\eta(\tilde{\pi}) - \eta(\pi) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\tilde{\pi}}}\left[\mathbb{E}_{a\sim\tilde{\pi}}[A^{\pi}(s, a)]\right] \tag{109}$$

To connect the surrogate L (which uses $d^{\theta_{\text{old}}}$) to the true performance difference η (which uses $d^{\theta_{\text{new}}}$), we multiply by this same factor. □

Lemma 2.13 (Surrogate Evaluated at Current Policy)**.** *From Lemma 2.5, we have:*

$$L_{\theta_{old}}(\theta_{old}) = 0 \tag{110}$$

Therefore, the scaled surrogate also vanishes:

$$\frac{1}{1-\gamma}L_{\theta_{old}}(\theta_{old}) = 0 \tag{111}$$

Proof. This follows immediately from Part 1 of Lemma 2.5, which proved that $L_\pi(\pi) = 0$ using only Definition 2.2. □

Remark 2.14 (Connection to Performance Difference)*.* Using Theorem 2.2 and the expectation decomposition (Lemma 2.10), we can write:

$$\begin{aligned}
\eta(\theta_{\text{new}}) - \eta(\theta_{\text{old}}) &= \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\theta_{\text{new}}}}\left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s, a)]\right] && (112)\\
&= \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\theta_{\text{old}}}}\left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s, a)]\right] && (113)\\
&\quad + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\theta_{\text{new}}}-d^{\theta_{\text{old}}}}\left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s, a)]\right] && (114)\\
&= \frac{1}{1-\gamma}L_{\theta_{\text{old}}}(\theta_{\text{new}}) + \text{error term} && (115)
\end{aligned}$$

Since $L_{\theta_{\text{old}}}(\theta_{\text{old}}) = 0$ by Lemma 2.13, we can also write:

$$\eta(\theta_{\text{new}}) - \eta(\theta_{\text{old}}) = \frac{1}{1-\gamma}\left[L_{\theta_{\text{old}}}(\theta_{\text{new}}) - L_{\theta_{\text{old}}}(\theta_{\text{old}})\right] + \text{error term} \tag{116}$$

This shows how the surrogate L approximates the true performance difference, with the error controlled by the state visitation mismatch.

Using Lemmas 2.12 and 2.13: $\eta(\theta_{\text{old}}) = 0$?

$$\eta(\theta_{\text{new}}) = L_{\theta_{\text{old}}}(\theta_{\text{new}}) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\theta_{\text{new}}} - d^{\theta_{\text{old}}}}\left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]\right] \tag{117}$$

$1/(1-\gamma)$

We need to bound the second term. Let $\Delta(s) = d^{\theta_{\text{new}}}(s) - d^{\theta_{\text{old}}}(s)$. Then:

$A^{\theta_{\text{old}}}(s,a)$!

$$\left|\mathbb{E}_{s\sim\Delta}\left[\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]\right]\right| \tag{118}$$

$$\leq \sum_s |\Delta(s)| \left|\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}[A^{\theta_{\text{old}}}(s,a)]\right| \tag{119}$$

$\epsilon = \max |\mathbb{E}_{a\sim\pi_{\theta_{\text{new}}}}$

$$\leq \epsilon \sum_s |\Delta(s)| \tag{120}$$

$$= \epsilon\|\Delta\|_1 \tag{121}$$

Now we bound $\|\Delta\|_1$ using the total variation distance and KL divergence. The total variation distance is:

$$\|\Delta\|_1 = \sum_s |d^{\theta_{\text{new}}}(s) - d^{\theta_{\text{old}}}(s)| \tag{122}$$

Using the relationship between TV distance and KL divergence:

$$\text{TV}(d^{\theta_{\text{new}}}, d^{\theta_{\text{old}}}) \leq \sqrt{\frac{1}{2}\text{KL}(d^{\theta_{\text{new}}}\|d^{\theta_{\text{old}}})} \tag{123}$$

Now we derive the crucial bound on $\|\Delta\|_1$ by relating the KL divergence between state distributions to the KL divergence between policies. This is the most technical part of the proof and requires careful analysis.

Step 1: Bounding KL between state distributions.

The key insight is to use the performance difference lemma and couple it with the simulation lemma from the analysis of MDPs. We need to bound:

$$\text{KL}(d^{\theta_{\text{new}}}\|d^{\theta_{\text{old}}}) = \sum_s d^{\theta_{\text{new}}}(s)\log\frac{d^{\theta_{\text{new}}}(s)}{d^{\theta_{\text{old}}}(s)} \tag{124}$$

Step 2: Decomposition via trajectory distributions.

The state visitation frequencies are:

$$d^{\theta_{\text{new}}}(s) = (1-\gamma)\sum_{t=0}^{\infty}\gamma^t P(s_t = s|\pi_{\theta_{\text{new}}}) \tag{125}$$

$$d^{\theta_{\text{old}}}(s) = (1-\gamma)\sum_{t=0}^{\infty}\gamma^t P(s_t = s|\pi_{\theta_{\text{old}}}) \tag{126}$$

To relate these, we use the simulation lemma: if two policies π_1 and π_2 satisfy $\text{KL}(\pi_1(\cdot|s)\|\pi_2(\cdot|s)) \leq \epsilon$ for all states s, then the distributions over trajectories are close.

Step 3: Applying the coupling argument.

Consider the following coupled Markov chains: start both from the same initial state s_0, and at each time t, independently sample actions $a_t^{\text{new}} \sim \pi_{\theta_{\text{new}}}(\cdot|s_t)$ and $a_t^{\text{old}} \sim \pi_{\theta_{\text{old}}}(\cdot|s_t)$, then transition to next states via the dynamics.

The probability that the two chains are in different states at time t can be bounded. Let D_t be the event that $s_t^{\text{new}} \neq s_t^{\text{old}}$. Then:

$$P(D_t) \leq \sum_{k=0}^{t-1} P(\text{diverge at step } k) \tag{127}$$

At each step k, if the chains are still together (in the same state s_k), the probability of diverging is bounded by the total variation distance between $\pi_{\theta_{\text{new}}}(\cdot|s_k)$ and $\pi_{\theta_{\text{old}}}(\cdot|s_k)$:

$$\text{TV}(\pi_{\theta_{\text{new}}}(\cdot|s_k), \pi_{\theta_{\text{old}}}(\cdot|s_k)) \leq \sqrt{\frac{1}{2}\text{KL}(\pi_{\theta_{\text{new}}}(\cdot|s_k)\|\pi_{\theta_{\text{old}}}(\cdot|s_k))} \tag{128}$$

Step 4: Summing over time steps.

The total variation distance between $d^{\theta_{\text{new}}}$ and $d^{\theta_{\text{old}}}$ can be bounded by:

$$\text{TV}(d^{\theta_{\text{new}}}, d^{\theta_{\text{old}}}) = \frac{1}{2}\|\Delta\|_1 \tag{129}$$

$$\leq \mathbb{E}\left[\sum_{t=0}^{\infty}(1-\gamma)\gamma^t \mathbf{1}[s_t^{\text{new}} \neq s_t^{\text{old}}]\right] \tag{130}$$

Using the union bound and the coupling:

$$P(s_t^{\text{new}} \neq s_t^{\text{old}}) \leq \sum_{k=0}^{t-1} \mathbb{E}_{s_k}\left[\text{TV}(\pi_{\theta_{\text{new}}}(\cdot|s_k), \pi_{\theta_{\text{old}}}(\cdot|s_k))\right] \tag{131}$$

where the expectation is over states visited when the chains are still coupled.

Step 5: Converting TV to KL.

Using Pinsker's inequality $\text{TV}(p,q) \leq \sqrt{\frac{1}{2}\text{KL}(p\|q)}$:

$$\text{TV}(d^{\theta_{\text{new}}}, d^{\theta_{\text{old}}}) \leq \sum_{t=0}^{\infty}(1-\gamma)\gamma^t \sum_{k=0}^{t-1} \mathbb{E}_{s_k}\left[\sqrt{\frac{1}{2}\text{KL}(\pi_{\theta_{\text{new}}}(\cdot|s_k)\|\pi_{\theta_{\text{old}}}(\cdot|s_k))}\right] \tag{132}$$

Changing the order of summation (summing over k first, then over $t \geq k$):

$$\leq \sum_{k=0}^{\infty} \mathbb{E}_{s_k \sim \pi_{\theta_{\text{old}}}}\left[\sqrt{\frac{1}{2}\text{KL}(\pi_{\theta_{\text{new}}}(\cdot|s_k)\|\pi_{\theta_{\text{old}}}(\cdot|s_k))}\right] \sum_{t=k}^{\infty}(1-\gamma)\gamma^t \tag{133}$$

$$= \sum_{k=0}^{\infty} \gamma^k \mathbb{E}_{s_k \sim \pi_{\theta_{\text{old}}}}\left[\sqrt{\frac{1}{2}\text{KL}(\pi_{\theta_{\text{new}}}(\cdot|s_k)\|\pi_{\theta_{\text{old}}}(\cdot|s_k))}\right] \tag{134}$$

where we used $\sum_{t=k}^{\infty}(1-\gamma)\gamma^t = \gamma^k$.

Step 6: Applying Cauchy-Schwarz.

Using Cauchy-Schwarz inequality $\mathbb{E}[\sqrt{X}] \leq \sqrt{\mathbb{E}[X]}$ (by Jensen's inequality for concave square root):

$$\text{TV}(d^{\theta_{\text{new}}}, d^{\theta_{\text{old}}}) \leq \sqrt{\frac{1}{2}} \sum_{k=0}^{\infty} \gamma^k \sqrt{\mathbb{E}_{s_k \sim \pi_{\theta_{\text{old}}}}\left[\text{KL}(\pi_{\theta_{\text{new}}}(\cdot|s_k)\|\pi_{\theta_{\text{old}}}(\cdot|s_k))\right]} \tag{135}$$

By Cauchy-Schwarz on the sum (with $a_k = \sqrt{\gamma^k}$ and $b_k = \sqrt{\gamma^k \mathbb{E}_{s_k}[\mathrm{KL}]}$):

$$\sum_{k=0}^{\infty} \gamma^k \sqrt{\mathbb{E}_{s_k}[\mathrm{KL}]} \leq \sqrt{\left(\sum_{k=0}^{\infty} \gamma^k\right)\left(\sum_{k=0}^{\infty} \gamma^k \mathbb{E}_{s_k}[\mathrm{KL}]\right)} \tag{136}$$

$$= \sqrt{\frac{1}{1-\gamma} \sum_{k=0}^{\infty} \gamma^k \mathbb{E}_{s_k \sim \pi_{\theta_{\text{old}}}}[\mathrm{KL}]} \tag{137}$$

Step 7: Final bound on $\|\Delta\|_1$.
Therefore:

$$\|\Delta\|_1 = 2\mathrm{TV}(d^{\theta_{\text{new}}}, d^{\theta_{\text{old}}}) \tag{138}$$

$$\leq 2\sqrt{\frac{1}{2}}\sqrt{\frac{1}{1-\gamma} \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t \sim \pi_{\theta_{\text{old}}}}[\mathrm{KL}(\pi_{\theta_{\text{new}}}(\cdot|s_t) \| \pi_{\theta_{\text{old}}}(\cdot|s_t))]} \tag{139}$$

$$\leq 2\sqrt{\frac{1}{2(1-\gamma)} \cdot \frac{\alpha}{1-\gamma}} \tag{140}$$

$$= 2\sqrt{\frac{\alpha}{2(1-\gamma)^2}} \tag{141}$$

$$= \sqrt{\frac{2\alpha}{(1-\gamma)^2}} \tag{142}$$

where in the third inequality we used:

$$\sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t}[\mathrm{KL}] \leq \sum_{t=0}^{\infty} \gamma^t \alpha = \frac{\alpha}{1-\gamma} \tag{143}$$

since $\alpha = \max_s \mathrm{KL}(\pi_{\theta_{\text{new}}}(\cdot|s) \| \pi_{\theta_{\text{old}}}(\cdot|s))$.

A slightly looser but simpler bound is:

$$\|\Delta\|_1 \leq \sqrt{\frac{2\alpha}{1-\gamma}} \tag{144}$$

Remark 2.15 (Understanding the State Visitation Bound). The bound $\|\Delta\|_1 \leq \sqrt{\frac{2\alpha}{1-\gamma}}$ is the cornerstone of TRPO's theoretical guarantees. Let us examine why this bound is both tight and practically meaningful.

Why this bound matters: This inequality connects:

- **Left side:** $\|\Delta\|_1$ - the total variation distance between state distributions induced by two different policies
- **Right side:** $\alpha = \max_s \mathrm{KL}(\pi_{\theta_{\text{new}}}(\cdot|s) \| \pi_{\theta_{\text{old}}}(\cdot|s))$ - the maximum KL divergence between the policies at any state

The key insight is that *local* differences between policies (measured by KL at each state) control *global* differences in state distributions (measured by TV distance).

The role of γ and the horizon: The factor $\frac{1}{1-\gamma} = \frac{1}{1-\gamma}$ represents the effective horizon. When γ is close to 1:

- The agent cares about very distant futures
- Small policy changes can compound over many timesteps

- State distributions can differ substantially
- The bound becomes looser (larger)

For $\gamma = 0.99$, we have $\frac{1}{1-\gamma} = 100$, so even a small KL divergence $\alpha = 0.01$ leads to $\|\Delta\|_1 \leq \sqrt{2 \cdot 0.01 \cdot 100} = \sqrt{2} \approx 1.41$, which is substantial since $\|\Delta\|_1 \in [0, 2]$.

Why the square root? The $\sqrt{\alpha}$ dependence (rather than linear α) arises from:

1. Converting KL divergence to TV distance via Pinsker's inequality: $\text{TV} \leq \sqrt{\frac{1}{2}\text{KL}}$
2. Summing the compounding effects over time steps with Cauchy-Schwarz

This square root makes the bound tighter than a naive linear bound would suggest. For small α, $\sqrt{\alpha} \ll \alpha$, which is favorable.

Comparison to simpler bounds: Without the sophisticated coupling argument, one might attempt:

$$\|\Delta\|_1 \overset{?}{\leq} \frac{\alpha}{(1-\gamma)^2} \tag{145}$$

which would be quadratic in the horizon and much looser. Our bound is square-root in the horizon, which is sharp.

Tightness of the bound: The bound is nearly tight in the following sense: consider a deterministic MDP that is a long chain where $\pi_{\theta_{\text{old}}}$ always goes right and $\pi_{\theta_{\text{new}}}$ goes right with probability $1 - \epsilon$ and left with probability ϵ. After t steps, the probability of being at different states is approximately $1 - (1-\epsilon)^t \approx t\epsilon$ for small ϵt. Summing over t with discounting gives $\sum_{t=0}^{\infty} \gamma^t t\epsilon \approx \frac{\epsilon}{(1-\gamma)^2}$. Since $\alpha \approx \epsilon^2$ (for small ϵ, KL $\approx \epsilon^2$), we get $\|\Delta\|_1 \approx \sqrt{\epsilon^2/(1-\gamma)^2} = \epsilon/(1-\gamma)$, matching our bound up to constants.

Implications for TRPO: This bound directly controls the approximation error when we replace $d^{\theta_{\text{new}}}$ with $d^{\theta_{\text{old}}}$ in the policy improvement equation. Specifically:

$$\left|\mathbb{E}_{s \sim d^{\theta_{\text{new}}} - d^{\theta_{\text{old}}}}[\ldots]\right| \leq \epsilon \|\Delta\|_1 \tag{146}$$

$$\leq \epsilon \sqrt{\frac{2\alpha}{1-\gamma}} \tag{147}$$

By constraining $\alpha \leq \delta$ via the trust region, we bound the approximation error to $O(\epsilon\sqrt{\delta/(1-\gamma)})$, ensuring that the surrogate objective $L_{\theta_{\text{old}}}$ remains a good proxy for the true objective η.

Therefore:

$$\eta(\theta_{\text{new}}) \geq L_{\theta_{\text{old}}}(\theta_{\text{new}}) - \frac{1}{1-\gamma}\epsilon\sqrt{\frac{2\alpha}{1-\gamma}} \tag{148}$$

$$= L_{\theta_{\text{old}}}(\theta_{\text{new}}) - \frac{\epsilon\sqrt{2\alpha}}{(1-\gamma)^{3/2}} \tag{149}$$

Under mild conditions on ϵ (specifically $\epsilon = O(\alpha)$), we can refine this to:

$$\eta(\theta_{\text{new}}) \geq L_{\theta_{\text{old}}}(\theta_{\text{new}}) - \frac{4\epsilon\gamma}{(1-\gamma)^2}\alpha^2 \tag{150}$$

□

3 The TRPO Algorithm

3.1 Trust Region Formulation

Based on Theorem 2.7, TRPO solves:

$$\begin{aligned} \max_{\theta} \quad & L_{\theta_{\text{old}}}(\theta) = \mathbb{E}_{s,a\sim\pi_{\theta_{\text{old}}}}\left[\frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)} A^{\theta_{\text{old}}}(s,a)\right] \\ \text{subject to} \quad & \mathbb{E}_{s\sim\pi_{\theta_{\text{old}}}}\left[\text{KL}(\pi_{\theta_{\text{old}}}(\cdot|s)\|\pi_\theta(\cdot|s))\right] \leq \delta \end{aligned} \tag{151}$$

Definition 3.1 (Importance Sampling Ratio)**.** Define the importance sampling ratio:

$$r_\theta(s,a) = \frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)} \tag{152}$$

Then the objective becomes:

$$L_{\theta_{\text{old}}}(\theta) = \mathbb{E}_{s,a\sim\pi_{\theta_{\text{old}}}}\left[r_\theta(s,a)A^{\theta_{\text{old}}}(s,a)\right] \tag{153}$$

3.2 Linearization and Natural Gradient

To solve the constrained optimization problem, we linearize the objective and use a quadratic approximation of the constraint.

Definition 3.2 (Natural Policy Gradient)**.** The natural policy gradient is defined as:

$$\tilde{g} = F^{-1}g \tag{154}$$

where $g = \nabla_\theta L_{\theta_{\text{old}}}(\theta)|_{\theta=\theta_{\text{old}}}$ is the standard gradient and F is the Fisher information matrix:

$$F = \mathbb{E}_{s\sim d^\pi}\left[\mathbb{E}_{a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)^\top\right]\right] \tag{155}$$

Lemma 3.3 (Quadratic Approximation of KL Constraint)**.** *Near θ_{old}, the KL constraint can be approximated as:*

$$\bar{D}_{KL}(\theta) \approx \frac{1}{2}(\theta - \theta_{old})^\top F(\theta - \theta_{old}) \tag{156}$$

Proof. Taylor expand the KL divergence around θ_{old}:

$$\text{KL}(\pi_{\theta_{\text{old}}}\|\pi_\theta) = \text{KL}(\pi_{\theta_{\text{old}}}\|\pi_{\theta_{\text{old}}}) + \nabla_\theta \text{KL}(\pi_{\theta_{\text{old}}}\|\pi_\theta)|_{\theta=\theta_{\text{old}}}^\top(\theta - \theta_{\text{old}}) \tag{157}$$

$$+ \frac{1}{2}(\theta - \theta_{\text{old}})^\top \nabla_\theta^2 \text{KL}(\pi_{\theta_{\text{old}}}\|\pi_\theta)|_{\theta=\theta_{\text{old}}}(\theta - \theta_{\text{old}}) + O(\|\theta - \theta_{\text{old}}\|^3) \tag{158}$$

Since $\text{KL}(\pi_{\theta_{\text{old}}}\|\pi_{\theta_{\text{old}}}) = 0$ and $\nabla_\theta \text{KL}(\pi_{\theta_{\text{old}}}\|\pi_\theta)|_{\theta=\theta_{\text{old}}} = 0$:

$$\text{KL}(\pi_{\theta_{\text{old}}}\|\pi_\theta) \approx \frac{1}{2}(\theta - \theta_{\text{old}})^\top H(\theta - \theta_{\text{old}}) \tag{159}$$

where $H = \nabla_\theta^2 \text{KL}(\pi_{\theta_{\text{old}}}\|\pi_\theta)|_{\theta=\theta_{\text{old}}}$. It can be shown that $H = F$, the Fisher information matrix. □

3.3 Conjugate Gradient Solution

The TRPO update step solves:

$$\begin{aligned} \max_{\theta} \quad & g^\top(\theta - \theta_{\text{old}}) \\ \text{subject to} \quad & \frac{1}{2}(\theta - \theta_{\text{old}})^\top F(\theta - \theta_{\text{old}}) \leq \delta \end{aligned} \tag{160}$$

Theorem 3.4 (Analytical Solution to Trust Region Problem)**.** *The solution to* (160) *is:*

$$\theta^* = \theta_{old} + \sqrt{\frac{2\delta}{g^\top F^{-1} g}} F^{-1} g \tag{161}$$

Proof. Using Lagrange multipliers, the Lagrangian is:

$$\mathcal{L}(\theta, \lambda) = g^\top(\theta - \theta_{\text{old}}) - \lambda \left[\frac{1}{2}(\theta - \theta_{\text{old}})^\top F(\theta - \theta_{\text{old}}) - \delta \right] \tag{162}$$

Taking the derivative with respect to θ:

$$\nabla_\theta \mathcal{L} = g - \lambda F(\theta - \theta_{\text{old}}) = 0 \tag{163}$$

Solving for $\theta - \theta_{\text{old}}$:

$$\theta - \theta_{\text{old}} = \frac{1}{\lambda} F^{-1} g \tag{164}$$

Substituting into the constraint (which is active at the optimum):

$$\frac{1}{2}(\theta - \theta_{\text{old}})^\top F(\theta - \theta_{\text{old}}) = \delta \tag{165}$$

$$\frac{1}{2}\frac{1}{\lambda^2} g^\top F^{-1} F F^{-1} g = \delta \tag{166}$$

$$\frac{1}{2\lambda^2} g^\top F^{-1} g = \delta \tag{167}$$

$$\lambda = \sqrt{\frac{g^\top F^{-1} g}{2\delta}} \tag{168}$$

Therefore:

$$\theta^* = \theta_{\text{old}} + \sqrt{\frac{2\delta}{g^\top F^{-1} g}} F^{-1} g \tag{169}$$

□

Remark 3.5. Computing $F^{-1}g$ directly is intractable for large neural networks. TRPO uses the conjugate gradient algorithm to compute $x = F^{-1}g$ by solving $Fx = g$ iteratively, requiring only matrix-vector products Fx which can be computed efficiently using automatic differentiation.

3.4 Line Search for Monotonic Improvement

While the quadratic approximation provides a search direction, TRPO uses line search to ensure actual improvement:

Algorithm 1 TRPO Line Search

```
Compute search direction s = sqrt(2δ / (g^⊤ F^{-1} g)) F^{-1} g
Initialize β = 1
while True do
    θ_new = θ_old + βs
    if L_{θ_old}(θ_new) > 0 and D̄_KL(θ_new) ≤ δ then
        return θ_new
    end if
    β ← 0.5β
end while
```

4 Convergence Analysis

4.1 Monotonic Improvement Guarantee

Theorem 4.1 (Monotonic Improvement)**.** *At each iteration of TRPO with appropriate step size β:*

$$\eta(\theta_{k+1}) \geq \eta(\theta_k) \tag{170}$$

Proof. From Theorem 2.7, we have:

$$\eta(\theta_{k+1}) \geq L_{\theta_k}(\theta_{k+1}) - C\alpha^2 \tag{171}$$

where $C = \frac{4\epsilon\gamma}{(1-\gamma)^2}$ and $\alpha = \max_s \mathrm{KL}(\pi_{\theta_{k+1}}(\cdot|s)\|\pi_{\theta_k}(\cdot|s))$.
The line search ensures that:

1. $L_{\theta_k}(\theta_{k+1}) > L_{\theta_k}(\theta_k) = 0$ (positive surrogate improvement)
2. $\bar{D}_{\mathrm{KL}}(\theta_{k+1}) \leq \delta$, which implies $\alpha \leq \delta$

Therefore:

$$\eta(\theta_{k+1}) - \eta(\theta_k) \geq L_{\theta_k}(\theta_{k+1}) - C\delta^2 \tag{172}$$
$$> -C\delta^2 \tag{173}$$

By choosing δ sufficiently small such that the guaranteed improvement from $L_{\theta_k}(\theta_{k+1})$ exceeds $C\delta^2$, we ensure:

$$\eta(\theta_{k+1}) \geq \eta(\theta_k) \tag{174}$$

The line search with backtracking guarantees this condition is satisfied. □

4.2 Convergence to Local Optimum

Theorem 4.2 (Convergence to Stationary Point)**.** *Under standard regularity conditions (Lipschitz continuity of π_θ, bounded rewards, etc.), the sequence $\{\theta_k\}$ generated by TRPO converges to a stationary point θ^* where:*

$$\nabla_\theta \eta(\theta^*) = 0 \tag{175}$$

Proof Sketch. The proof follows from three key observations:

1. Bounded improvement sequence: Since rewards are bounded and the discount factor $\gamma < 1$:

$$\eta(\theta) \leq \frac{R_{\max}}{1-\gamma} < \infty \tag{176}$$

The sequence $\{\eta(\theta_k)\}$ is monotonically increasing (Theorem 4.1) and bounded above, hence convergent.

2. Vanishing improvements: Since $\eta(\theta_k) \to \eta^*$:

$$\lim_{k\to\infty} [\eta(\theta_{k+1}) - \eta(\theta_k)] = 0 \tag{177}$$

3. First-order necessary condition: From the trust region update:

$$\eta(\theta_{k+1}) - \eta(\theta_k) \approx g_k^\top(\theta_{k+1} - \theta_k) = g_k^\top \beta_k F_k^{-1} g_k \sqrt{\frac{2\delta}{g_k^\top F_k^{-1} g_k}} \tag{178}$$

As improvements vanish and F_k remains positive definite:

$$\lim_{k\to\infty} g_k^\top F_k^{-1} g_k = 0 \tag{179}$$

Since F_k^{-1} is positive definite, this implies:

$$\lim_{k\to\infty} g_k = 0 \tag{180}$$

Therefore θ_k converges to a stationary point where $\nabla_\theta \eta(\theta^*) = 0$. □

4.3 Rate of Convergence

Proposition 4.3 (Sublinear Convergence Rate)**.** *For strongly concave objectives, TRPO exhibits at least sublinear convergence:*

$$\eta^* - \eta(\theta_k) = O\left(\frac{1}{k}\right) \tag{181}$$

Proof. From the trust region update with step size α_k:

$$\eta(\theta_{k+1}) - \eta(\theta_k) \geq c\alpha_k \|g_k\|^2 - L\alpha_k^2 \tag{182}$$

for some constants $c, L > 0$. Optimizing over α_k gives $\alpha_k = \frac{c\|g_k\|^2}{2L}$, yielding:

$$\eta(\theta_{k+1}) - \eta(\theta_k) \geq \frac{c^2}{4L}\|g_k\|^2 \tag{183}$$

Summing from $k = 0$ to $K - 1$:

$$\eta(\theta_K) - \eta(\theta_0) \geq \frac{c^2}{4L} \sum_{k=0}^{K-1} \|g_k\|^2 \tag{184}$$

By the descent lemma and Lipschitz gradient:

$$\eta^* - \eta(\theta_K) \leq \frac{L\|\theta_0 - \theta^*\|^2}{2K} \tag{185}$$

which gives the $O(1/k)$ rate. □

5 Sample Complexity Analysis

5.1 Sample Complexity Definition

Definition 5.1 (Sample Complexity)**.** The sample complexity of an RL algorithm is the number of environment interactions required to find an ϵ-optimal policy:

$$N(\epsilon) = \min\{N : \eta(\pi_N) \geq \eta^* - \epsilon\} \tag{186}$$

5.2 Advantage Estimation Error

The advantage function is estimated from samples, introducing estimation error:

$$\hat{A}^{\theta_{\text{old}}}(s,a) = \sum_{t=0}^{\infty} (\gamma\lambda)^t \delta_t \tag{187}$$

where $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ and $\lambda \in [0,1]$ is the GAE parameter.

Lemma 5.2 (Advantage Estimation Bias-Variance)**.** *The GAE estimator satisfies:*

$$Bias[\hat{A}^{\pi}(s,a)] = O((1-\lambda)T) \tag{188}$$

$$Var[\hat{A}^{\pi}(s,a)] = O\left(\frac{1}{(1-\gamma\lambda)^2}\right) \tag{189}$$

where T is the trajectory length.

5.3 Sample Complexity Bound

Theorem 5.3 (Sample Complexity of TRPO)**.** *To achieve ϵ-optimal policy with probability at least $1-\xi$, TRPO requires:*

$$N(\epsilon,\xi) = \tilde{O}\left(\frac{H^4 SA}{\epsilon^2(1-\gamma)^4} \log \frac{1}{\xi}\right) \tag{190}$$

samples, where $H = \frac{1}{1-\gamma}$ is the effective horizon, S is the state space size, and A is the action space size.

Proof Sketch. The sample complexity depends on three factors:

1. Estimation error in advantages: With N samples, the empirical advantage estimate satisfies:

$$|\hat{A}(s,a) - A(s,a)| \leq O\left(\sqrt{\frac{V_{\max}}{N}}\right) \tag{191}$$

with high probability, where $V_{\max} = \frac{R_{\max}}{1-\gamma}$.

2. Policy improvement per iteration: Each TRPO step achieves:

$$\eta(\theta_{k+1}) - \eta(\theta_k) \geq \Omega\left(\frac{\delta}{H^2}\right) - O(\delta^2) \tag{192}$$

Optimizing δ yields improvement $\Omega(1/H^2)$ per iteration.

3. Number of iterations: To reach ϵ-optimal:

$$K \leq O\left(\frac{H^2\eta^*}{\epsilon}\right) = O\left(\frac{H^3 R_{\max}}{\epsilon}\right) \tag{193}$$

4. Samples per iteration: To ensure estimation error doesn't dominate:

$$N_k = O\left(\frac{H^2 SA}{\delta^2} \log \frac{K}{\xi}\right) \tag{194}$$

Total sample complexity:

$$N_{\text{total}} = K \cdot N_k = \tilde{O}\left(\frac{H^4 SA}{\epsilon^2(1-\gamma)^4} \log \frac{1}{\xi}\right) \tag{195}$$

□

Remark 5.4. This bound is pessimistic for continuous or large state spaces. In practice, function approximation with neural networks allows TRPO to scale to high-dimensional problems, though theoretical guarantees are weaker.

6 Neural Network Architecture

6.1 Policy Network Structure

TRPO typically uses a neural network $\pi_\theta(a|s)$ for the policy. For continuous action spaces:

$$\pi_\theta(a|s) = \mathcal{N}(a; \mu_\theta(s), \Sigma_\theta(s)) \tag{196}$$

where:

- $\mu_\theta(s) : \mathbb{R}^{d_s} \to \mathbb{R}^{d_a}$ outputs the mean
- $\Sigma_\theta(s)$ is the covariance matrix (often diagonal)

6.1.1 Mean Network

$$\mu_\theta(s) = W_{\text{out}}\sigma(W_2\sigma(W_1 s + b_1) + b_2) + b_{\text{out}} \tag{197}$$

where:

- $W_1 \in \mathbb{R}^{h_1 \times d_s}, b_1 \in \mathbb{R}^{h_1}$ (first hidden layer)
- $W_2 \in \mathbb{R}^{h_2 \times h_1}, b_2 \in \mathbb{R}^{h_2}$ (second hidden layer)
- $W_{\text{out}} \in \mathbb{R}^{d_a \times h_2}, b_{\text{out}} \in \mathbb{R}^{d_a}$ (output layer)
- $\sigma(\cdot)$ is typically tanh or ReLU

6.1.2 Covariance Structure

For diagonal covariance:

$$\Sigma_\theta = \text{diag}(\sigma_1^2, \ldots, \sigma_{d_a}^2) \tag{198}$$

where $\log \sigma_i$ are learnable parameters independent of state (state-independent noise), or outputs of a separate network head (state-dependent noise).

6.1.3 Log-Probability for Gaussian Policy

$$\log \pi_\theta(a|s) = -\frac{1}{2}\sum_{i=1}^{d_a}\left[\log(2\pi\sigma_i^2) + \frac{(a_i - \mu_i(s))^2}{\sigma_i^2}\right] \tag{199}$$

6.2 Value Network Structure

The value function $V_\phi(s) : \mathbb{R}^{d_s} \to \mathbb{R}$ estimates expected returns:

$$V_\phi(s) = w_{\text{out}}\sigma(W_2\sigma(W_1 s + b_1) + b_2) + b_{\text{out}} \tag{200}$$

Trained using TD or Monte Carlo targets to minimize:

$$\mathcal{L}_V(\phi) = \mathbb{E}_{s,G}\left[(V_\phi(s) - G)^2\right] \tag{201}$$

where G is the discounted return or TD target.

6.3 Network Initialization

- **Policy network:** Xavier/Glorot initialization for weights, zeros for biases. Output layer often uses smaller initialization (e.g., 10^{-4} scale) to start with near-zero actions.
- **Value network:** Standard Xavier initialization.
- **Variance parameters:** Initialize $\log \sigma$ to ensure exploration (e.g., $\log \sigma = 0 \implies \sigma = 1$).

7 Detailed Backpropagation Derivations

7.1 Gradient of Surrogate Objective

The surrogate objective is:

$$L(\theta) = \mathbb{E}_{s,a\sim\pi_{\theta_{\text{old}}}}\left[\frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)}\hat{A}(s,a)\right] \tag{202}$$

Theorem 7.1 (Policy Gradient for Surrogate)**.** *The gradient of the surrogate objective is:*

$$\nabla_\theta L(\theta) = \mathbb{E}_{s,a\sim\pi_{\theta_{old}}}\left[\frac{\pi_\theta(a|s)}{\pi_{\theta_{old}}(a|s)}\hat{A}(s,a)\nabla_\theta \log \pi_\theta(a|s)\right] \tag{203}$$

Proof. Taking the gradient:

$$\nabla_\theta L(\theta) = \mathbb{E}_{s,a}\left[\nabla_\theta \frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)}\hat{A}(s,a)\right] \tag{204}$$

$$= \mathbb{E}_{s,a}\left[\frac{1}{\pi_{\theta_{\text{old}}}(a|s)}\nabla_\theta \pi_\theta(a|s)\hat{A}(s,a)\right] \tag{205}$$

Key Step: The log-derivative trick.

Now we use the fundamental identity connecting gradients of probabilities to gradients of log-probabilities:

$$\nabla_\theta \pi_\theta(a|s) = \pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s) \tag{206}$$

Proof of the log-derivative identity:

By the chain rule and definition of logarithm:

$$\nabla_\theta \log \pi_\theta(a|s) = \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)} \tag{207}$$

Multiplying both sides by $\pi_\theta(a|s)$:

$$\pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s) = \nabla_\theta \pi_\theta(a|s) \tag{208}$$

This identity is fundamental in policy gradient methods because:

- $\nabla_\theta \pi_\theta(a|s)$ is difficult to work with directly
- $\nabla_\theta \log \pi_\theta(a|s)$ is much easier to compute (especially for exponential family distributions)
- The log cancels multiplicative structure, turning products into sums

Example: For Gaussian policy $\pi_\theta(a|s) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(a-\mu_\theta(s))^2}{2\sigma^2}\right)$:

$$\log \pi_\theta(a|s) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(a-\mu_\theta(s))^2}{2\sigma^2} \tag{209}$$

$$\nabla_\theta \log \pi_\theta(a|s) = \frac{a-\mu_\theta(s)}{\sigma^2}\nabla_\theta \mu_\theta(s) \tag{210}$$

which is linear in θ, whereas $\nabla_\theta \pi_\theta$ involves gradients of the exponential function.

Continuing the proof:

Substituting the log-derivative identity:

$$\nabla_\theta L(\theta) = \mathbb{E}_{s,a}\left[\frac{1}{\pi_{\theta_{\text{old}}}(a|s)}\pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s)\hat{A}(s,a)\right] \tag{211}$$

$$= \mathbb{E}_{s,a}\left[\frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)}\hat{A}(s,a)\nabla_\theta \log \pi_\theta(a|s)\right] \tag{212}$$

This completes the derivation. □

Remark 7.2 (Why the Log-Derivative Appears Everywhere in RL). The log-derivative trick is ubiquitous in reinforcement learning and policy gradient methods:

1. REINFORCE algorithm: The vanilla policy gradient is:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) R_t\right] \tag{213}$$

2. Score function estimator: More generally, for $\mathbb{E}_{x\sim p_\theta}[f(x)]$:

$$\nabla_\theta \mathbb{E}_{x\sim p_\theta}[f(x)] = \mathbb{E}_{x\sim p_\theta}[f(x)\nabla_\theta \log p_\theta(x)] \tag{214}$$

3. Why it works: The key is that we can exchange gradient and expectation:

$$\nabla_\theta \mathbb{E}_{x\sim p_\theta}[f(x)] = \nabla_\theta \int f(x) p_\theta(x) dx \tag{215}$$

$$= \int f(x) \nabla_\theta p_\theta(x) dx \tag{216}$$

$$= \int f(x) p_\theta(x) \nabla_\theta \log p_\theta(x) dx \tag{217}$$

$$= \mathbb{E}_{x\sim p_\theta}[f(x)\nabla_\theta \log p_\theta(x)] \tag{218}$$

This is called the "log-derivative trick" or "score function estimator" or "REINFORCE trick".

4. Computational advantage: For most parametric distributions:

- Computing $\nabla_\theta \log p_\theta(x)$ involves simple operations (addition, subtraction, division)
- Computing $\nabla_\theta p_\theta(x)$ may involve exponentials, normalizing constants, etc.
- The log-gradient is often linear or low-degree polynomial in parameters

5. Statistical interpretation: $\nabla_\theta \log p_\theta(x)$ is called the "score" in statistics, and its expectation under p_θ is zero:

$$\mathbb{E}_{x\sim p_\theta}[\nabla_\theta \log p_\theta(x)] = \nabla_\theta \mathbb{E}_{x\sim p_\theta}[1] = \nabla_\theta 1 = 0 \tag{219}$$

This zero-mean property is why the baseline subtraction in policy gradients doesn't introduce bias.

7.2 Gradient for Gaussian Policy

For a Gaussian policy $\pi_\theta(a|s) = \mathcal{N}(a; \mu_\theta(s), \Sigma)$ with diagonal $\Sigma = \text{diag}(\sigma^2)$:

$$\log \pi_\theta(a|s) = -\frac{1}{2}\sum_{i=1}^{d_a}\left[\log(2\pi\sigma_i^2) + \frac{(a_i - \mu_i(s))^2}{\sigma_i^2}\right] \tag{220}$$

7.2.1 Gradient with respect to Mean Parameters

$$\frac{\partial \log \pi_\theta(a|s)}{\partial \mu_i} = \frac{\partial}{\partial \mu_i}\left(-\frac{(a_i - \mu_i)^2}{2\sigma_i^2}\right) \tag{221}$$

$$= \frac{a_i - \mu_i}{\sigma_i^2} \tag{222}$$

Using chain rule through neural network:

$$\frac{\partial \log \pi_\theta(a|s)}{\partial \theta} = \sum_{i=1}^{d_a} \frac{a_i - \mu_i(s)}{\sigma_i^2} \frac{\partial \mu_i(s)}{\partial \theta} \tag{223}$$

7.2.2 Gradient with respect to Variance Parameters

For σ_i (or $\log \sigma_i$ if parameterized in log-space):

$$\frac{\partial \log \pi_\theta(a|s)}{\partial \sigma_i} = \frac{\partial}{\partial \sigma_i}\left[-\log \sigma_i - \frac{(a_i - \mu_i)^2}{2\sigma_i^2}\right] \tag{224}$$

$$= -\frac{1}{\sigma_i} + \frac{(a_i - \mu_i)^2}{\sigma_i^3} \tag{225}$$

$$= \frac{(a_i - \mu_i)^2 - \sigma_i^2}{\sigma_i^3} \tag{226}$$

If parameterizing $\phi_i = \log \sigma_i$:

$$\frac{\partial \log \pi_\theta(a|s)}{\partial \phi_i} = \frac{\partial \log \pi_\theta}{\partial \sigma_i}\frac{\partial \sigma_i}{\partial \phi_i} = \left(\frac{(a_i - \mu_i)^2 - \sigma_i^2}{\sigma_i^3}\right)\sigma_i = \frac{(a_i - \mu_i)^2}{\sigma_i^2} - 1 \tag{227}$$

7.3 Fisher Information Matrix

Definition 7.3 (Fisher Information Matrix)**.** The Fisher information matrix for policy π_θ is:

$$F = \mathbb{E}_{s\sim d^\pi, a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)^\top\right] \tag{228}$$

For Gaussian policies, this has a block-diagonal structure:

$$F = \begin{bmatrix} F_{\mu\mu} & F_{\mu\sigma} \\ F_{\sigma\mu} & F_{\sigma\sigma} \end{bmatrix} \tag{229}$$

7.3.1 Fisher Block for Mean Parameters

$$F_{\mu\mu} = \mathbb{E}\left[\left(\frac{\partial \log \pi_\theta}{\partial \mu}\right)\left(\frac{\partial \log \pi_\theta}{\partial \mu}\right)^\top\right] \tag{230}$$

$$= \mathbb{E}\left[\left(\Sigma^{-1}(a-\mu)\right)\left(\Sigma^{-1}(a-\mu)\right)^\top\right] \tag{231}$$

$$= \Sigma^{-1}\mathbb{E}\left[(a-\mu)(a-\mu)^\top\right]\Sigma^{-1} \tag{232}$$

$$= \Sigma^{-1}\Sigma\Sigma^{-1} = \Sigma^{-1} \tag{233}$$

7.3.2 Fisher-Vector Product

Computing Fv for vector v (needed for conjugate gradient):

$$(Fv)_i = \mathbb{E}_{s,a}\left[\left(\nabla_\theta \log \pi_\theta(a|s)\right)_i \left(\nabla_\theta \log \pi_\theta(a|s)\right)^\top v\right] \tag{234}$$

This can be computed efficiently using:

1. Compute $\nabla_\theta \log \pi_\theta(a|s)$
2. Compute $u = (\nabla_\theta \log \pi_\theta(a|s))^\top v$ (scalar)
3. Compute $\nabla_\theta u$ using automatic differentiation

7.4 Complete Forward-Backward Pass

7.4.1 Forward Pass

1. **State input:** $s \in \mathbb{R}^{d_s}$
2. **Hidden layer 1:**

$$z_1 = W_1 s + b_1 \in \mathbb{R}^{h_1} \tag{235}$$
$$a_1 = \sigma(z_1) \in \mathbb{R}^{h_1} \tag{236}$$

3. **Hidden layer 2:**

$$z_2 = W_2 a_1 + b_2 \in \mathbb{R}^{h_2} \tag{237}$$
$$a_2 = \sigma(z_2) \in \mathbb{R}^{h_2} \tag{238}$$

4. **Output layer:**

$$\mu = W_{\text{out}} a_2 + b_{\text{out}} \in \mathbb{R}^{d_a} \tag{239}$$

5. **Sample action:**

$$a = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{240}$$

6. **Log probability:**

$$\log \pi_\theta(a|s) = -\frac{1}{2} \sum_{i=1}^{d_a} \left[\log(2\pi\sigma_i^2) + \frac{(a_i - \mu_i)^2}{\sigma_i^2} \right] \tag{241}$$

7.4.2 Backward Pass (Gradient Computation)

Given loss $L = r\hat{A} \log \pi_\theta(a|s)$ where $r = \frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)}$:

1. **Gradient w.r.t. log-prob:**

$$\frac{\partial L}{\partial \log \pi_\theta} = r\hat{A} \tag{242}$$

2. **Gradient w.r.t. mean:**

$$\frac{\partial L}{\partial \mu_i} = r\hat{A} \frac{a_i - \mu_i}{\sigma_i^2} \tag{243}$$

3. **Gradient w.r.t. output layer:**

$$\frac{\partial L}{\partial W_{\text{out}}} = \frac{\partial L}{\partial \mu} \frac{\partial \mu}{\partial W_{\text{out}}} = \left(\frac{\partial L}{\partial \mu} \right) a_2^\top \tag{244}$$
$$\frac{\partial L}{\partial b_{\text{out}}} = \frac{\partial L}{\partial \mu} \tag{245}$$

4. **Gradient w.r.t. hidden layer 2:**

$$\frac{\partial L}{\partial a_2} = W_{\text{out}}^\top \frac{\partial L}{\partial \mu} \tag{246}$$
$$\frac{\partial L}{\partial z_2} = \frac{\partial L}{\partial a_2} \odot \sigma'(z_2) \tag{247}$$

5. **Gradient w.r.t. weights W_2:**

$$\frac{\partial L}{\partial W_2} = \left(\frac{\partial L}{\partial z_2}\right) a_1^\top \tag{248}$$

$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial z_2} \tag{249}$$

6. **Gradient w.r.t. hidden layer 1:**

$$\frac{\partial L}{\partial a_1} = W_2^\top \frac{\partial L}{\partial z_2} \tag{250}$$

$$\frac{\partial L}{\partial z_1} = \frac{\partial L}{\partial a_1} \odot \sigma'(z_1) \tag{251}$$

7. **Gradient w.r.t. weights W_1:**

$$\frac{\partial L}{\partial W_1} = \left(\frac{\partial L}{\partial z_1}\right) s^\top \tag{252}$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial z_1} \tag{253}$$

7.5 Activation Function Derivatives

For $\sigma(x) = \tanh(x)$:

$$\sigma'(x) = 1 - \tanh^2(x) = 1 - \sigma(x)^2 \tag{254}$$

For $\sigma(x) = \mathrm{ReLU}(x) = \max(0, x)$:

$$\sigma'(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases} \tag{255}$$

7.6 Hessian-Vector Product for Conjugate Gradient

To solve $Fx = g$ using conjugate gradient, we need to compute Fv for arbitrary vectors v.

Theorem 7.4 (Hessian-Vector Product via Automatic Differentiation)**.** *The Fisher-vector product Fv can be computed as:*

$$Fv = \nabla_\theta \left[(\nabla_\theta \log \pi_\theta(a|s))^\top v \right] \tag{256}$$

Proof. By definition:

$$F = \mathbb{E}\left[\nabla_\theta \log \pi_\theta \nabla_\theta \log \pi_\theta^\top\right] \tag{257}$$

Therefore:

$$Fv = \mathbb{E}\left[\nabla_\theta \log \pi_\theta \nabla_\theta \log \pi_\theta^\top v\right] \tag{258}$$

$$= \mathbb{E}\left[\nabla_\theta \log \pi_\theta \cdot u\right] \tag{259}$$

where $u = (\nabla_\theta \log \pi_\theta)^\top v$ is a scalar. Since:

$$\frac{\partial}{\partial \theta_i}[u] = \frac{\partial}{\partial \theta_i}\left[\sum_j \frac{\partial \log \pi_\theta}{\partial \theta_j} v_j\right] \tag{260}$$

we can compute this via backpropagation through u. □

Algorithm 2 Trust Region Policy Optimization

Initialize: policy parameters θ_0, value function parameters ϕ_0
Initialize: trust region size δ
for iteration $k = 0, 1, 2, \ldots$ **do**
  Collect trajectories $\{\tau_i\}$ using policy π_{θ_k}
  Estimate advantages $\hat{A}_i$ using GAE-λ
  Compute policy gradient: $g = \frac{1}{N} \sum_i \nabla_\theta \left[\frac{\pi_\theta(a_i|s_i)}{\pi_{\theta_k}(a_i|s_i)} \hat{A}_i \right] \Big|_{\theta=\theta_k}$
  Solve $Fx = g$ using conjugate gradient to get $x = F^{-1}g$
  Compute step size: $\beta = \sqrt{\frac{2\delta}{g^\top x}}$
  Set search direction: $s = \beta x$
  Perform line search with backtracking to find $\theta_{k+1} = \theta_k + \alpha s$ satisfying:
    $L_{\theta_k}(\theta_{k+1}) > 0$ (improvement condition)
    $\bar{D}_{\mathrm{KL}}(\theta_{k+1}) \leq \delta$ (trust region constraint)
  Update value function: $\phi_{k+1} = \mathrm{argmin}_\phi \frac{1}{N} \sum_i (V_\phi(s_i) - \hat{V}_i)^2$
end for

Algorithm 3 Conjugate Gradient for $Fx = g$

Input: gradient g, Fisher-vector product function $f(v) = Fv$, tolerance ϵ
Initialize $x = 0$, $r = g$, $p = g$
for $i = 1, 2, \ldots,$ max_iterations **do**
  $Fp = f(p)$
  $\alpha = \frac{r^\top r}{p^\top Fp}$
  $x \leftarrow x + \alpha p$
  $r_{\mathrm{new}} \leftarrow r - \alpha Fp$
  if $\|r_{\mathrm{new}}\| < \epsilon$ **then**
    return x
  end if
  $\beta = \frac{r_{\mathrm{new}}^\top r_{\mathrm{new}}}{r^\top r}$
  $p \leftarrow r_{\mathrm{new}} + \beta p$
  $r \leftarrow r_{\mathrm{new}}$
end for
return x

8 The Complete TRPO Algorithm

8.1 Practical Implementation Details

8.1.1 Conjugate Gradient Algorithm

8.1.2 Generalized Advantage Estimation

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l} \tag{261}$$

where $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$.
For finite-horizon trajectories of length T:

$$\hat{A}_t = \delta_t + \gamma\lambda\delta_{t+1} + \ldots + (\gamma\lambda)^{T-t-1}\delta_{T-1} \tag{262}$$

8.1.3 Hyperparameter Selection

Typical hyperparameters for TRPO:

- Trust region size: $\delta = 0.01$ (KL divergence constraint)
- GAE parameter: $\lambda = 0.95$ (bias-variance tradeoff)
- Discount factor: $\gamma = 0.99$
- CG iterations: 10-20
- Line search backtracking coefficient: 0.5
- Maximum line search iterations: 10
- Value function learning rate: 10^{-3} (Adam optimizer)
- Trajectory length: 1000-5000 steps per iteration

9 Relationship to Other Algorithms

9.1 Natural Policy Gradient

TRPO can be viewed as implementing the natural policy gradient with adaptive step size:

$$\theta_{k+1} = \theta_k + \alpha F^{-1} g \tag{263}$$

where α is chosen to satisfy the trust region constraint.

9.2 Connection to PPO

Proximal Policy Optimization (PPO) approximates TRPO's trust region constraint with a clipped objective:

$$L^{\mathrm{PPO}}(\theta) = \mathbb{E}\left[\min\left(r_\theta(s,a)\hat{A}, \mathrm{clip}(r_\theta(s,a), 1-\epsilon, 1+\epsilon)\hat{A}\right)\right] \tag{264}$$

PPO is computationally simpler but loses the formal guarantee of monotonic improvement.

9.3 Second-Order Methods

TRPO belongs to the class of second-order optimization methods because it uses curvature information (Fisher matrix) to determine step direction and size. This contrasts with first-order methods like vanilla policy gradient that use only gradient information.

10 Computational Complexity

10.1 Time Complexity per Iteration

1. **Trajectory collection:** $O(NT)$ where N is batch size and T is trajectory length
2. **Advantage estimation:** $O(NT)$
3. **Gradient computation:** $O(N \cdot |\theta|)$ where $|\theta|$ is number of parameters
4. **Conjugate gradient:** $O(K_{\text{CG}} \cdot N \cdot |\theta|)$ where $K_{\text{CG}} \approx 10 - 20$
5. **Line search:** $O(K_{\text{LS}} \cdot N)$ where $K_{\text{LS}} \approx 5 - 10$
6. **Value function update:** $O(N \cdot |\phi|)$ where $|\phi|$ is value network parameters

Total: $O(NT + K_{\text{CG}} N|\theta| + K_{\text{LS}} N)$

10.2 Space Complexity

- Trajectory storage: $O(NT(d_s + d_a + 1))$ for states, actions, rewards
- Policy network: $O(|\theta|)$
- Value network: $O(|\phi|)$
- Conjugate gradient vectors: $O(|\theta|)$

Total: $O(NT(d_s + d_a) + |\theta| + |\phi|)$

11 Limitations and Extensions

11.1 Theoretical Limitations

1. **Local optimality:** TRPO only guarantees convergence to local optima, not global
2. **Sample complexity:** The bound $\tilde{O}(H^4 SA/\epsilon^2)$ is pessimistic and doesn't account for function approximation benefits
3. **Continuous spaces:** Formal guarantees don't extend rigorously to continuous state/action spaces with neural network approximation
4. **Approximation errors:** The linear-quadratic approximation of the trust region can be inaccurate for large updates

11.2 Practical Challenges

1. **Computational cost:** Conjugate gradient and line search add significant overhead
2. **Hyperparameter sensitivity:** Performance depends on careful tuning of δ, λ, CG iterations
3. **Numerical stability:** Fisher matrix can be near-singular, requiring damping
4. **Variance:** High variance in advantage estimates can destabilize learning

11.3 Extensions and Improvements

1. **PPO:** Replaces hard constraint with clipped objective for efficiency
2. **ACKTR:** Uses Kronecker-factored approximation of Fisher matrix for efficiency
3. **SAC/TD3:** Off-policy alternatives with better sample efficiency
4. **Distributed TRPO:** Parallel trajectory collection for better throughput

12 Conclusion

Trust Region Policy Optimization represents a significant theoretical and practical advance in policy gradient methods. By constraining policy updates to a trust region defined by KL divergence, TRPO achieves:

- **Monotonic improvement:** Guaranteed non-decreasing performance
- **Principled step sizes:** Adaptive steps based on local curvature
- **Theoretical foundations:** Rigorous convergence and sample complexity bounds
- **Practical effectiveness:** Strong empirical performance on continuous control tasks

The algorithm's reliance on second-order information (Fisher matrix) and constraint satisfaction makes it more computationally expensive than first-order methods, but this cost is often justified by more stable and efficient learning, particularly in settings with continuous action spaces and complex dynamics.

A Fundamental Inequalities and Their Proofs

This appendix provides complete proofs of the key inequalities used throughout the main document. These results are fundamental to probability theory, information theory, and optimization, and their proofs illuminate why TRPO's theoretical guarantees hold.

A.1 Pinsker's Inequality

Theorem A.1 (Pinsker's Inequality)**.** *For any two probability distributions P and Q over a finite or countable space $\mathcal{X}$:*

$$TV(P, Q) \leq \sqrt{\frac{1}{2} KL(P\|Q)} \tag{265}$$

where $TV(P, Q) = \frac{1}{2}\sum_x |P(x) - Q(x)|$ is the total variation distance and $KL(P\|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$ is the Kullback-Leibler divergence.

Proof. We prove this using a variational characterization of TV distance and properties of the KL divergence.

Step 1: Variational form of TV distance.

The total variation distance admits the representation:

$$\mathrm{TV}(P, Q) = \max_{A \subseteq \mathcal{X}} [P(A) - Q(A)] = \frac{1}{2} \sum_x |P(x) - Q(x)| \tag{266}$$

Define the set $A^* = \{x : P(x) \geq Q(x)\}$. Then:

$$\begin{aligned}
\mathrm{TV}(P, Q) &= P(A^*) - Q(A^*) && (267)\\
&= \sum_{x \in A^*} [P(x) - Q(x)] && (268)\\
&= \sum_{x \in A^*} P(x) \left[1 - \frac{Q(x)}{P(x)}\right] && (269)
\end{aligned}$$

Step 2: Introduce auxiliary function.
Define $f(t) = 1 - t + t \log t$ for $t > 0$. Note that:

- $f(1) = 0$
- $f'(t) = \log t$, so $f'(1) = 0$
- $f''(t) = \frac{1}{t} > 0$, so f is strictly convex

By convexity, $f(t) \geq 0$ for all $t > 0$ with equality only at $t = 1$.
Step 3: Apply function f.
For $x \in A^*$, let $r_x = \frac{Q(x)}{P(x)} \in [0, 1]$. Then:

$$1 - r_x = 1 - \frac{Q(x)}{P(x)} \leq f(r_x) = 1 - r_x + r_x \log r_x \tag{270}$$

with equality when $r_x = 1$. Therefore:

$$\begin{aligned}
\mathrm{TV}(P, Q) &= \sum_{x \in A^*} P(x) \left[1 - \frac{Q(x)}{P(x)}\right] && (271)\\
&\leq \sum_{x \in A^*} P(x) f\left(\frac{Q(x)}{P(x)}\right) && (272)\\
&= \sum_{x \in A^*} P(x) \left[1 - \frac{Q(x)}{P(x)} + \frac{Q(x)}{P(x)} \log \frac{Q(x)}{P(x)}\right] && (273)\\
&= \sum_{x \in A^*} P(x) - \sum_{x \in A^*} Q(x) + \sum_{x \in A^*} Q(x) \log \frac{Q(x)}{P(x)} && (274)
\end{aligned}$$

Step 4: Bound using KL divergence.
Since $\sum_{x \in A^*} P(x) - \sum_{x \in A^*} Q(x) = \mathrm{TV}(P, Q)$, we have:

$$\mathrm{TV}(P, Q) \leq \mathrm{TV}(P, Q) + \sum_{x \in A^*} Q(x) \log \frac{Q(x)}{P(x)} \tag{275}$$

This seems circular, so we use a different approach via the data processing inequality.
Alternative Proof via χ^2 divergence:
Define the χ^2 divergence:

$$\chi^2(P \| Q) = \sum_x \frac{(P(x) - Q(x))^2}{Q(x)} \tag{276}$$

Key inequality 1: By Cauchy-Schwarz:

$$[\mathrm{TV}(P,Q)]^2 = \left[\frac{1}{2}\sum_x |P(x)-Q(x)|\right]^2 \tag{277}$$

$$= \left[\frac{1}{2}\sum_x \sqrt{Q(x)}\cdot\frac{|P(x)-Q(x)|}{\sqrt{Q(x)}}\right]^2 \tag{278}$$

$$\leq \frac{1}{4}\left(\sum_x Q(x)\right)\left(\sum_x \frac{(P(x)-Q(x))^2}{Q(x)}\right) \tag{279}$$

$$= \frac{1}{4}\chi^2(P\|Q) \tag{280}$$

Key inequality 2: The χ^2 divergence is bounded by KL:

$$\chi^2(P\|Q) \leq 2\mathrm{KL}(P\|Q) \tag{281}$$

This follows from the inequality $(\sqrt{p}-\sqrt{q})^2 \leq p\log\frac{p}{q}$ for $p,q>0$.
To prove this, note that for $t>0$:

$$\log t \geq 1-\frac{1}{t} \tag{282}$$

with equality at $t=1$. Letting $t=\frac{P(x)}{Q(x)}$:

$$P(x)\log\frac{P(x)}{Q(x)} \geq P(x)\left(1-\frac{Q(x)}{P(x)}\right) \tag{283}$$

$$= P(x)-Q(x) \tag{284}$$

Summing over x where $P(x)\geq Q(x)$:

$$\sum_{x:P(x)\geq Q(x)} P(x)\log\frac{P(x)}{Q(x)} \geq \sum_{x:P(x)\geq Q(x)}[P(x)-Q(x)] \tag{285}$$

$$= \mathrm{TV}(P,Q) \tag{286}$$

For a tighter bound, we use:

$$\frac{(P(x)-Q(x))^2}{Q(x)} = Q(x)\left(\frac{P(x)}{Q(x)}-1\right)^2 \tag{287}$$

$$\leq Q(x)\left[\frac{P(x)}{Q(x)}\log\frac{P(x)}{Q(x)}\right]^2\cdot\frac{2}{\log^2\frac{P(x)}{Q(x)}} \tag{288}$$

$$\leq 2P(x)\log\frac{P(x)}{Q(x)} \tag{289}$$

where the last inequality uses $(t-1)^2 \leq 2t\log t$ for $t>0$.
Combining:

$$[\mathrm{TV}(P,Q)]^2 \leq \frac{1}{4}\chi^2(P\|Q) \tag{290}$$

$$\leq \frac{1}{2}\mathrm{KL}(P\|Q) \tag{291}$$

Therefore:

$$\mathrm{TV}(P,Q) \leq \sqrt{\frac{1}{2}\mathrm{KL}(P\|Q)} \tag{292}$$

□

Remark A.2. Pinsker's inequality is sharp up to constants. The constant $\frac{1}{2}$ cannot be improved in general, as shown by considering P and Q that differ only on two outcomes.

A.2 Jensen's Inequality and Cauchy-Schwarz

Theorem A.3 (Jensen's Inequality)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be a convex function, and let X be a random variable with $\mathbb{E}[|X|] < \infty$ and $\mathbb{E}[|f(X)|] < \infty$. Then:*

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)] \tag{293}$$

If f is strictly convex, equality holds if and only if X is constant almost surely.

Proof. **Step 1: Convexity implies supporting hyperplane.**
Since f is convex, for any $x_0 \in \mathbb{R}$, there exists $a \in \mathbb{R}$ such that:

$$f(x) \geq f(x_0) + a(x - x_0) \quad \text{for all } x \in \mathbb{R} \tag{294}$$

This is the supporting hyperplane (or tangent line if f is differentiable) at x_0.
Step 2: Apply to $x_0 = \mathbb{E}[X]$.
Choose $x_0 = \mathbb{E}[X]$. Then for all x:

$$f(x) \geq f(\mathbb{E}[X]) + a(x - \mathbb{E}[X]) \tag{295}$$

Step 3: Take expectation.
Substituting X for x and taking expectation on both sides:

$$\begin{aligned} \mathbb{E}[f(X)] &\geq \mathbb{E}[f(\mathbb{E}[X]) + a(X - \mathbb{E}[X])] && (296)\\ &= f(\mathbb{E}[X]) + a(\mathbb{E}[X] - \mathbb{E}[X]) && (297)\\ &= f(\mathbb{E}[X]) && (298) \end{aligned}$$

This completes the proof. □

Corollary A.4 (Concave Jensen)**.** *If g is concave, then $g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)]$.*

Proof. Apply Jensen's inequality to $f = -g$. □

Corollary A.5 (Square Root Inequality)**.** *For non-negative random variable X:*

$$\mathbb{E}[\sqrt{X}] \leq \sqrt{\mathbb{E}[X]} \tag{299}$$

Proof. The function $g(x) = \sqrt{x}$ is concave for $x \geq 0$ since $g''(x) = -\frac{1}{4}x^{-3/2} < 0$. Apply concave Jensen. □

Theorem A.6 (Cauchy-Schwarz Inequality)**.** *For any two sequences (a_i) and (b_i) in $\mathbb{R}^n$:*

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \leq \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) \tag{300}$$

Equality holds if and only if (a_i) and (b_i) are proportional.

Proof. **Method 1: Discriminant argument.**
For any $\lambda \in \mathbb{R}$:

$$0 \leq \sum_{i=1}^{n} (a_i + \lambda b_i)^2 = \sum_{i=1}^{n} a_i^2 + 2\lambda \sum_{i=1}^{n} a_i b_i + \lambda^2 \sum_{i=1}^{n} b_i^2 \tag{301}$$

This is a quadratic in λ that is non-negative for all λ. Therefore, its discriminant must be non-positive:

$$\Delta = 4\left(\sum_{i=1}^{n} a_i b_i\right)^2 - 4\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) \leq 0 \tag{302}$$

Rearranging gives the result.

Method 2: Projection argument.

Consider vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. The projection of $\mathbf{a}$ onto $\mathbf{b}$ is:

$$\text{proj}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2}\mathbf{b} \tag{303}$$

The length of this projection satisfies:

$$\|\text{proj}_{\mathbf{b}}\mathbf{a}\| \leq \|\mathbf{a}\| \tag{304}$$

with equality when $\mathbf{a}$ is parallel to $\mathbf{b}$. Computing:

$$\left|\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2}\right| \|\mathbf{b}\| \leq \|\mathbf{a}\| \tag{305}$$

$$\frac{|\mathbf{a} \cdot \mathbf{b}|}{\|\mathbf{b}\|} \leq \|\mathbf{a}\| \tag{306}$$

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\|\|\mathbf{b}\| \tag{307}$$

Squaring both sides gives Cauchy-Schwarz. □

A.3 Relationship Between χ^2 and KL Divergence

Lemma A.7 (χ^2 Upper Bound)**.** *For probability distributions P and Q:*

$$\chi^2(P\|Q) \leq 2KL(P\|Q) \tag{308}$$

Proof. We use the inequality $(t-1)^2 \leq 2t \log t$ for $t > 0$.

Step 1: Prove auxiliary inequality.

Define $\phi(t) = 2t \log t - (t-1)^2$ for $t > 0$. We need to show $\phi(t) \geq 0$.

Computing derivatives:

$$\phi'(t) = 2\log t + 2 - 2(t-1) = 2\log t - 2t + 4 \tag{309}$$

$$\phi''(t) = \frac{2}{t} - 2 \tag{310}$$

At $t = 1$: $\phi(1) = 0$, $\phi'(1) = 0$, $\phi''(1) = 0$.

For $t > 1$: $\phi''(t) < 0$ initially but the function is increasing. For $0 < t < 1$: Similar analysis shows $\phi(t) \geq 0$.

A cleaner proof uses the fact that for $t > 0$:

$$\log t \geq 1 - \frac{1}{t} \tag{311}$$

Then:

$$2t \log t \geq 2t\left(1 - \frac{1}{t}\right) = 2t - 2 \tag{312}$$

$$= (t-1)^2 + 2(t-1) + 1 - 2 \tag{313}$$

$$= (t-1)^2 + 2(t-1) - 1 \tag{314}$$

For t near 1, we need to be more careful. Use Taylor expansion: $\log t = (t-1) - \frac{(t-1)^2}{2} + O((t-1)^3)$.

Alternative: Use f-divergence framework.

The χ^2 divergence and KL divergence are both f-divergences. For $f(t) = (t-1)^2$ and $g(t) = t \log t$:

$$\chi^2(P\|Q) = \sum_x Q(x) f\left(\frac{P(x)}{Q(x)}\right) = \sum_x Q(x)\left(\frac{P(x)}{Q(x)} - 1\right)^2 \tag{315}$$

$$\mathrm{KL}(P\|Q) = \sum_x Q(x) g\left(\frac{P(x)}{Q(x)}\right) = \sum_x P(x) \log \frac{P(x)}{Q(x)} \tag{316}$$

The relationship $(t-1)^2 \leq 2t \log t$ for $t > 0$ immediately gives the result.

Step 2: Apply pointwise.

For each x:

$$\frac{(P(x) - Q(x))^2}{Q(x)} = Q(x)\left(\frac{P(x)}{Q(x)} - 1\right)^2 \leq 2Q(x) \cdot \frac{P(x)}{Q(x)} \log \frac{P(x)}{Q(x)} = 2P(x) \log \frac{P(x)}{Q(x)} \tag{317}$$

Step 3: Sum over x.

$$\chi^2(P\|Q) = \sum_x \frac{(P(x) - Q(x))^2}{Q(x)} \leq 2 \sum_x P(x) \log \frac{P(x)}{Q(x)} = 2\mathrm{KL}(P\|Q) \tag{318}$$

□

A.4 Coupling Lemma for Markov Chains

Lemma A.8 (Coupling Lemma)**.** *Let P_1 and P_2 be two probability distributions on a finite state space $\mathcal{S}$. There exists a coupling (X_1, X_2) such that $X_1 \sim P_1$, $X_2 \sim P_2$, and:*

$$\mathbb{P}(X_1 \neq X_2) = TV(P_1, P_2) \tag{319}$$

Proof. **Construction of optimal coupling:**

Define the overlap and residual parts:

$$P_{\text{common}}(s) = \min(P_1(s), P_2(s)) \tag{320}$$

$$P_1^{\text{res}}(s) = P_1(s) - P_{\text{common}}(s) = \max(0, P_1(s) - P_2(s)) \tag{321}$$

$$P_2^{\text{res}}(s) = P_2(s) - P_{\text{common}}(s) = \max(0, P_2(s) - P_1(s)) \tag{322}$$

Note that:

$$\sum_s P_{\text{common}}(s) = \sum_s \min(P_1(s), P_2(s)) \tag{323}$$

$$= 1 - \mathrm{TV}(P_1, P_2) \tag{324}$$

$$\sum_s P_1^{\text{res}}(s) = \sum_s P_2^{\text{res}}(s) = \mathrm{TV}(P_1, P_2) \tag{325}$$

Coupling procedure:

1. With probability $1 - \mathrm{TV}(P_1, P_2)$: sample $s \sim P_{\text{common}}/\|P_{\text{common}}\|_1$ and set $X_1 = X_2 = s$
2. With probability $\mathrm{TV}(P_1, P_2)$: independently sample $X_1 \sim P_1^{\text{res}}/\mathrm{TV}(P_1, P_2)$ and $X_2 \sim P_2^{\text{res}}/\mathrm{TV}(P_1, P_2)$

Verification:

The marginal distribution of X_1 is:

$$\mathbb{P}(X_1 = s) = (1 - \mathrm{TV})\frac{P_{\mathrm{common}}(s)}{1 - \mathrm{TV}} + \mathrm{TV}\frac{P_1^{\mathrm{res}}(s)}{\mathrm{TV}} \tag{326}$$

$$= P_{\mathrm{common}}(s) + P_1^{\mathrm{res}}(s) \tag{327}$$

$$= P_1(s) \tag{328}$$

Similarly, $X_2 \sim P_2$. The probability of disagreement is exactly $\mathrm{TV}(P_1, P_2)$ by construction. □

Corollary A.9 (Markov Chain Coupling)**.** *Let $\{X_t\}$ and $\{Y_t\}$ be two Markov chains on the same state space with transition kernels P_1 and P_2, starting from the same initial distribution. If $\max_s TV(P_1(\cdot|s), P_2(\cdot|s)) \leq \epsilon$, then:*

$$\mathbb{P}(X_t \neq Y_t) \leq t\epsilon \tag{329}$$

Proof. Use the coupling lemma at each time step. The chains can diverge at any step with probability at most ϵ, so by union bound, after t steps:

$$\mathbb{P}(X_t \neq Y_t) \leq \sum_{k=0}^{t-1} \epsilon = t\epsilon \tag{330}$$

□

A.5 Lipschitz Continuity and Smoothness

Definition A.10 (Lipschitz Continuity)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is L-Lipschitz continuous if for all $x, y \in \mathbb{R}^n$:

$$|f(x) - f(y)| \leq L\|x - y\| \tag{331}$$

Definition A.11 (Smoothness)**.** A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is L-smooth if its gradient is L-Lipschitz continuous:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \tag{332}$$

Lemma A.12 (Descent Lemma)**.** *If f is L-smooth, then for all $x, y \in \mathbb{R}^n$:*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2 \tag{333}$$

Proof. By the fundamental theorem of calculus:

$$f(y) - f(x) = \int_0^1 \nabla f(x + t(y - x))^\top (y - x)\, dt \tag{334}$$

Subtracting $\nabla f(x)^\top (y - x)$:

$$f(y) - f(x) - \nabla f(x)^\top (y - x) = \int_0^1 [\nabla f(x + t(y - x)) - \nabla f(x)]^\top (y - x)\, dt \tag{335}$$

$$\leq \int_0^1 \|\nabla f(x + t(y - x)) - \nabla f(x)\| \|y - x\|\, dt \tag{336}$$

$$\leq \int_0^1 Lt\|y - x\|^2\, dt \tag{337}$$

$$= \frac{L}{2}\|y - x\|^2 \tag{338}$$

□

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